"What Were They Thinking?" Using Open-Text Responses to Validate Constructs in Survey Experiments

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Abstract

Conjoint experiments are the dominant survey design in political science, yet they rest on an assumption that is rarely tested: that respondents reason about profile attributes in the way the construct of interest specifies. Open-text questions are already familiar as basic manipulation checks, but they can do much more. I argue they should be taken seriously as evidence about construct validity and experimental design, and I develop a framework that combines structural topic modeling (STM) for unsupervised discovery with large language model (LLM) classification for theory-driven confirmation. I demonstrate the framework across three conjoint experiments. The primary application, a conjoint on immigrant naturalization in South Korea and Taiwan, shows that a social integration cue shifts respondent reasoning from economic and practical criteria toward civic engagement, with effects of 7–18 percentage points across both methods and six LLM classifiers (Fleiss' $\kappa = 0.745$). A second application, a standard conjoint on immigrant admission, shows the framework transfers to designs without within-attribute randomization and can also refine constructs that an earlier analysis grouped too coarsely. A third application, a conjoint on North Korean migrant entrepreneurship with expert human-coded classifications, benchmarks LLM performance against that ground truth and shows that iterative codebook refinement produces substantial LLM-human agreement. I also extract token-level log-probabilities from the classification step to measure model uncertainty, an approach not used in text-as-data workflows. These probabilities predict when models disagree but reveal that all models are overconfident relative to human coding.

Keywords: survey experiments, construct validation, open-ended questions, structural topic models, large language models, text classification, immigration attitudes

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Introduction

Conjoint experiments are the most widely used survey design in contemporary political science (Hainmueller et al., 2014). Respondents evaluate multi-attribute profiles and make forced choices, producing average marginal component effects (AMCEs) for each attribute. But conjoint designs carry an untested assumption: that respondents reason about profile attributes in the way the construct of interest specifies, and that closed-ended response items capture that reasoning. A conjoint profile may contain ten or more randomly varied attributes, yet researchers typically interpret treatment effects as if respondents attended to and processed each attribute independently. The possibility that respondents focused on entirely different aspects of the profile, or interpreted attributes in unexpected ways, is rarely examined directly.

I propose embedding open-text questions in conjoint experiments and using computational text analysis to examine respondent reasoning directly. When a respondent is asked "Why did you choose this person?" after a conjoint task, the answer reveals the reasoning behind the observed choice. Open-text responses are already familiar as basic manipulation checks, verifying that respondents noticed or encoded an experimental stimulus (Ziegler, 2022). My claim is stronger: open-text responses should be used as direct evidence about whether the construct the researcher intends to measure is in fact the construct respondents reason about, and about whether the experimental design activates that reasoning. That is a question about construct validity and about the experimental design itself, not a check on stimulus delivery. I demonstrate the approach across three conjoint surveys, each representing a distinct validation scenario: a primary conjoint with within-attribute treatment randomization, which allows clean identification of how an experimental cue shifts reasoning; a standard conjoint without within-attribute randomization, which is the more common design in the literature and offers a second validation case based on retrospective construct refinement; and a conjoint for which expert human-coded classifications are available, which allows a third validation based on direct calibration of LLM performance against a trusted benchmark.

The analytical framework combines unsupervised and supervised text analysis. Structural topic modeling (STM) serves as a *discovery* tool: what topics did respondents discuss, and how did experimental

conditions affect topic prevalence (Roberts et al., 2014)? Large language model (LLM) classification provides *confirmation*: do open-text responses align with theory-derived categories, following recent codebook design frameworks (Halterman and Keith, 2025)? A third layer takes advantage of what makes LLMs different from human coders. The existing literature treats LLM output as text: the model generates a classification label, and the researcher evaluates it the same way they would evaluate a human coder's label (Törnberg, 2025; Heseltine and Clemm von Hohenberg, 2024; Chae and Davidson, 2025). But language models also produce a probability distribution over tokens at each generation step. Extracting the token-level log-probability (logprob) assigned to the classification label gives a measure of model uncertainty that is distinct from the self-reported confidence the model generates as text. I use these logprobs to assess whether the models are *calibrated*: does the predicted probability correspond to actual classification accuracy?

The primary application uses conjoint survey data on immigrant naturalization in South Korea and Taiwan, collected in 2024 (Hainmueller and Hopkins, 2015; Denney and Green, 2021). The conjoint includes a "social integration" attribute that randomly describes immigrants' civic engagement activities. The validation question is direct: when respondents see the social integration cue, do they write about civic engagement? They do. The cue increases civic reasoning by 7–18 percentage points while reducing economic and credential-based reasoning, across both methods and six LLM classifiers (Fleiss' $\kappa = 0.745$). The second application extends the framework to Denney and Green (2021), an earlier conjoint on immigrant admission preferences in South Korea. Because that design lacks within-attribute treatment randomization, it tests whether the framework transfers to the more common setup; it also shows that open-text evidence can decompose a construct that the original study grouped into a single broad category, refining measurement in a completed study. The third application turns to Denney et al. (2024), a conjoint on North Korean migrant entrepreneurship policy for which two trained human coders classified every open-text response. That dataset supplies the expert ground truth needed to calibrate LLM classifications directly, and it provides an independent test of whether the iterative codebook workflow produces meaningful gains in classification accuracy.

The contribution has three parts. First, I develop a two-stage workflow for construct validation in

conjoint experiments: unsupervised topic modeling discovers the empirical structure of open-text responses, and those findings inform the design of a theory-driven codebook for LLM classification. Unlike approaches that apply LLM classification or scaling directly (Mellon et al., 2024; DiGiuseppe and Flynn, 2026), this workflow uses the unsupervised step to check and refine the codebook before classification begins. Second, I show that token-level log-probabilities provide a useful measure of classification uncertainty. These probabilities predict inter-rater agreement ($r_s = 0.46$) but also reveal that all six models are overconfident relative to human-coded ground truth (Expected Calibration Error 0.15–0.26), which means researchers should not take LLM confidence at face value. Third, all classification uses six models (one proprietary, five open-weight from 3B to 72B parameters) to test reproducibility across model families and sizes (Barrie et al., 2025; Chae and Davidson, 2025).

Background and Related Work

Open-Ended Questions in Survey Research

Open-ended questions have a long history in survey research, but their systematic use for construct validation is less established. Geer (1988) showed that open-ended responses capture genuine attitudes rather than mere articulateness. Lupia (2018) documented persistent quality problems in coding open-ended responses and called for greater transparency. Glazier et al. (2021) introduced a "self-coding" method where respondents categorize their own open-ended responses, a direct check on whether researcher-imposed categories match respondent interpretations. Ziegler (2022) uses text-as-data methods on open-ended survey responses as manipulation checks. Haaland et al. (2025) review the growing use of open-ended survey data to uncover mechanisms behind beliefs and behavior, including LLM-based annotation at scale.

This paper extends that line of work by using computational text analysis to scale from small hand-coded samples to entire survey populations.

Structural Topic Models

Topic models provide a data-driven approach to discovering the thematic structure of text corpora (see Grimmer and Stewart, 2013; Grimmer et al., 2022; Gentzkow et al., 2019).[1] Roberts et al. (2014) developed the Structural Topic Model for contexts where document-level metadata (such as experimental treatment conditions) should inform topic estimation; Roberts et al. (2016) provide the formal statistical foundation. Unlike standard LDA, STM allows covariates to affect both topic prevalence (which topics appear in which documents) and topic content (which words are associated with which topics). This makes STM ideal for survey experiments: researchers can directly estimate how experimental conditions affect the topics respondents discuss (see also Egami et al., 2022; Fong and Grimmer, 2023, on causal inference with text).

Hobbs and Green (2025) distinguish between topic discovery and attitude inference, arguing that topic models excel at the former but require careful interpretation for the latter. The present application is primarily concerned with topic discovery (identifying what respondents discussed) and with treatment effects on topic prevalence.

LLM-Based Text Classification

Large language models are now widely used for text classification (Gilardi et al., 2023; Ziems et al., 2024; Bail, 2024). Chae and Davidson (2025) compare zero-shot, few-shot, and instruction-tuned LLMs for text classification in social science, finding that model size and prompting strategy jointly determine classification quality. The present study tests this directly by deploying models from 3B to 72B parameters. Törnberg (2025) demonstrates that LLMs can outperform human coders on contextual reasoning tasks. Heseltine and Clemm von Hohenberg (2024) show that hybrid human-LLM workflows achieve 93%+ accuracy cost-effectively. Ornstein et al. (2025) provide the `prompt` R package for few-shot LLM classification. Mellon et al. (2024) demonstrate that LLMs can code open-text survey responses at scale with accuracy comparable to human coders, while Rathje et al. (2024) show that GPT performs well for text analysis across 12 languages, supporting multilingual classification. Most of this work assigns each

[1] For the foundational Latent Dirichlet Allocation model, see Blei et al. (2003).

response to a discrete code. DiGiuseppe and Flynn (2026) take a different tack, using LLM pairwise comparisons and Bradley-Terry modeling to place responses on a continuous scale. The choice between categorical and continuous measurement depends on the research question; the present study asks what *kind* of reasoning respondents use, so categorical codes are the natural fit.

Codebook design matters more than model choice. Halterman and Keith (2025) argue that the quality of LLM classification depends on the precision of the codebook: loose instructions lead to poor performance regardless of model capability. They recommend codebooks with explicit definitions, inclusion criteria, exclusion criteria, and worked examples.

Reproducibility is an equally pressing concern. Barrie et al. (2025) show that proprietary LLM APIs exhibit substantial variance across runs and model versions, threatening the reproducibility of research findings. They recommend open-weight models that can be run locally with fixed parameters, producing deterministic outputs (see also Le Mens and Gallego, 2025). This concern motivates the present study's design: five open-weight models form the reproducible core, while GPT-4o-mini is included as a reference point for the proprietary models that dominate current practice in political science. Gilardi et al. (2023) used GPT-3.5-turbo, Törnberg (2025) and Heseltine and Clemm von Hohenberg (2024) used GPT-4, and Chae and Davidson (2025) used GPT-4o. The open-weight results show that researchers need not rely on proprietary APIs to achieve comparable classification quality.

Data and Research Design

Primary Application: Naturalization Conjoint

The primary data come from original surveys conducted in South Korea and Taiwan between January and June 2024, examining public attitudes toward immigrant naturalization (Denney et al., 2025). Respondents were recruited via Qualtrics using stratified quota sampling by sex, age, and geographic region; the final samples comprise $N = 1{,}999$ in South Korea and $N = 2{,}050$ in Taiwan. Each respondent evaluated paired immigrant profiles in a forced-choice conjoint (Hainmueller et al., 2014; Bansak et al., 2021), with profiles varying on 12 randomly assigned attributes (Supplementary Information, Table A.2).

The attribute of interest is *social integration*, which describes the immigrant's civic engagement with four levels ranging from "only associates with fellow expatriates" to "volunteers weekly at a local community center." This attribute was randomly withheld from some profiles, creating a clean cue-shown vs. cue-not-shown comparison indexed by the treatment variable `cue_shown`. Shares are approximately balanced by design (South Korea: 49.8% cue-shown; Taiwan: 49.1%).

After the conjoint task, respondents answered an open-text question: *"Why did you choose this person as more deserving of citizenship?"* The corpus comprises 4,049 responses in Korean and Traditional Chinese. Responses are brief (median 6–10 characters) but substantive: approximately 79% receive a substantive code under majority-vote LLM classification, which means they contain identifiable references to profile attributes.

Secondary Applications

Two earlier conjoints extend the analysis. Denney and Green (2021) is a 2020 conjoint on immigrant admission in South Korea ($N = 1{,}008$) with seven attributes and no within-attribute randomization, which provides a test of whether the framework transfers to the more common conjoint setup and whether open-text evidence can refine constructs an earlier analysis grouped too coarsely. Denney et al. (2024) is a 2020 conjoint on South Korean attitudes toward North Korean migrant entrepreneurship policy ($N = 2{,}010$) for which two trained human coders independently classified every open-text response (inter-coder $\kappa = 0.881$), providing the expert benchmark needed to calibrate LLM classifications and to test whether iterative codebook refinement closes the gap between LLM and expert coding. Table 1 summarizes all three.

	Naturalization	**Admission**	**N.K. Entrepreneurship**
Country	South Korea, Taiwan	South Korea	South Korea
N (respondents)	4,049	1,008	2,010
Year	2024	2020	2020
Source	Denney, Fraser & Steinhardt	Denney & Green (2021)	Denney, Ward & Green (2023)
Topic	Immigrant citizenship	Immigrant admission	N.K. migrant business support
Treatment	Civic cue randomized	None (standard conjoint)	None (standard conjoint)
Open-text language	Korean, Traditional Chinese	Korean (+ English translation)	Korean (+ English translation)
Classification codes	5	5	4
Human ground truth	No	No	Yes (2 coders, $\kappa = 0.881$)
Role in paper	Primary analysis	Framework portability	Calibration & validation

Table 1: Three conjoint datasets used in this paper. Note: N.K. = North Korea.

Analytical Framework

Approach 1: Structural Topic Modeling

I fit separate STM models for each country, using the `stm` package in R (Roberts et al., 2019) with spectral initialization and seed = 42. Korean text is preprocessed using the `kiwipiepy` morphological analyzer and Chinese text using `quanteda`'s built-in segmenter (Benoit et al., 2018). Preprocessing follows established guidelines for multilingual text analysis (Lucas et al., 2015; Denny and Spirling, 2018).

Since the conjoint design is fully randomized, the prevalence specification includes only the treatment indicator:

$$\text{prevalence} \sim \texttt{cue_shown} \tag{1}$$

I select the number of topics (K) using the coherence-exclusivity tradeoff from `searchK()` diagnostics, resulting in $K = 6$ for South Korea and $K = 8$ for Taiwan.

Treatment effects on topic prevalence are estimated using `estimateEffect()`, which provides coefficient estimates, standard errors, and confidence intervals for the effect of `cue_shown` on each topic's prevalence.

Approach 2: LLM Classification

I developed the codebook iteratively, using STM findings to guide refinement. The initial version (v1) used four codes: `civic_commitment`, `identity_concern`, `economic_focus`, and `other`. The civic commitment code captured civic engagement, volunteering, and community involvement, but also included language competence and length of residence as signals of integration commitment.

Running v1 across GPT-4o-mini exposed a problem: 561 responses (13.9%) were coded as civic commitment based solely on language fluency or residence duration, which are practical attributes, not active civic engagement. The STM results showed that topics about language and residence *decrease* when the social integration cue is shown, meaning these attributes behave as substitutes for civic engagement, not signals of it. Including them under civic commitment conflated participation in community life with individual-level adaptation to the host country.

This STM-informed diagnosis led to a revised codebook (v2) with five codes:

Code	Definition
`civic_commitment`	Participation in community life: volunteering, community organizing, social participation. Outward-facing, prosocial engagement
`functional_integration`	Individual-level adaptation: language competence, residence duration, law-abiding behavior. Practical prerequisites for functioning in the host country
`identity_concern`	Ethnic, cultural, or national identity concerns, including religion and ethnicity when used as evaluation criteria
`economic_focus`	Economic contributions: jobs, taxes, career, occupation, work experience
`other`	Truly uninformative responses: filler words, gibberish, meta-comments. Brevity alone is never a reason for this code

Table 2: LLM classification codebook (v2). Codes follow the design principles of Halterman and Keith (2025), with explicit definitions, inclusion/exclusion criteria, and few-shot examples in both Korean and Traditional Chinese.

The classification pipeline uses six LLMs. GPT-4o-mini (OpenAI API) provides a reference point for the proprietary models used in prior work. Five open-weight models form the reproducible core:

Model	Parameters	Infrastructure	Role
GPT-4o-mini	—	OpenAI API	Proprietary baseline; representative of models used in current practice
Qwen 2.5 3B	3B	Consumer GPU	Open-weight; smallest model
Llama 3.1 8B	8B	Consumer GPU	Open-weight; mid-range
Gemma 3 12B	12B	University HPC (A100)	Open-weight; mid-range
Qwen 2.5 32B	32B	University HPC (A100)	Open-weight; large
Qwen 2.5 72B	72B	University HPC (A100)	Open-weight; largest model; size scaling

Table 3: LLM classifiers used in the analysis. Open-weight models are run via the Ollama framework with temperature = 0 and seed = 42 for deterministic output.

All open-weight models are run with temperature = 0, seed = 42, and max_tokens = 200, producing deterministic outputs. GPT-4o-mini is run with the same parameters, but proprietary API outputs are not strictly reproducible across runs due to batched inference and silent model updates (Barrie et al.,

2025). The open-weight models therefore provide the reproducibility guarantee; GPT-40-mini serves as a reference point for the proprietary models that dominate current practice. The range of model sizes (3B to 72B parameters, including three from the Qwen 2.5 family) allows a direct test of whether classification quality scales with model size (Chae and Davidson, 2025). The classification prompt also asks each model to self-report a confidence level (HIGH, MEDIUM, or LOW). This is what researchers encounter in normal LLM interaction: the model generates a confidence label as text, producing what it judges a plausible human coder would report. But this self-assessed confidence is itself a generated token with no necessary relationship to the model's actual predictive uncertainty.

LLMs are text-prediction machines, not human coders, and this difference can be put to use. A language model generates text one token at a time. At each step, it computes a probability distribution over its entire vocabulary (over 100,000 possible tokens) given all preceding text.[2] For each classification, I extract the log-probability (logprob) that the model assigns to the first token of the classification label. This reflects the model's predictive distribution at the moment of classification, not a self-assessment generated as text. Although the probability is computed over the full vocabulary, the classification prompt concentrates virtually all probability mass onto the five valid code prefixes; for the typical response, they capture over 99% of the distribution (Supplementary Information, Table **??**). The Supplementary Information reports calibration analysis showing how these token probabilities relate to inter-rater agreement and to accuracy against human-coded ground truth.

The classification pipeline is implemented in R and Python. Codebook generation (from a structured YAML definition following Halterman and Keith 2025), cross-model validation, and all statistical analyses are written in R; the OpenAI API classifier is a standalone Python script. Open-weight models are deployed through Ollama, a lightweight framework for running LLMs locally with fixed inference parameters. The two smallest models (3B, 8B) run on a consumer GPU (NVIDIA RTX 3060 Ti, 8GB

[2] The model's final layer produces a raw score (logit) for each token in the vocabulary. A softmax function converts these scores into a probability distribution summing to 1.0. The log-probability (logprob) is the natural logarithm of this probability. When the model generates the classification label after `"code": "`, the logprob of the chosen token (e.g., `civic`) reflects how much probability mass the model assigned to that classification relative to every other token it could have produced. If `civic` receives probability 0.92, the model has placed 92% of its predictive mass on that single classification, with the other four codes and the remaining 100,000+ tokens sharing 8%. If instead the probability is 0.45 with `functional` at 0.35, the model is actually torn between two classifications, even though it still selects `civic` and might self-report HIGH confidence in both cases.

VRAM); the three larger models (12B, 32B, 72B) run on a university high-performance computing cluster equipped with NVIDIA A100 GPUs, with the 32B and 72B models quantized to 4-bit precision to fit within GPU memory. All code, codebooks, and classification outputs are available in the replication archive.

Each response is classified independently by all six models. Because no human-coded ground truth exists for these responses, the analysis relies on cross-model consensus rather than any single model's judgment. The majority-vote classification assigns each response the code that four or more of the six models agree on; responses without a four-model consensus (7.2% of the corpus) are excluded. This approach treats the six LLMs as diverse raters (varying in architecture, size, and training data, though not fully independent given shared pretraining corpora) and applies inter-rater reliability standards conventional in content analysis (Benoit et al., 2026). Because the models share pretraining data, agreement statistics likely overstate what fully independent annotators would produce. The ground-truth extension, where LLM-human agreement reaches $\kappa \approx 0.6$–0.7, provides a more realistic lower bound on classification quality than the inter-model κ of 0.745. Majority-vote aggregation reduces the influence of model-specific biases (for example, Llama 3.1 8B's inflated functional integration rate in Taiwan) by requiring broad agreement across model families and sizes before any classification is assigned. Treatment effects are then estimated as differences in code prevalence between cue-shown and cue-not-shown conditions on this consensus-classified corpus, with bootstrap confidence intervals (2,000 iterations, percentile method) providing inference.

The individual models produce consistent results even without aggregation. Table C.6 in the Supplementary Information reports the cue effect for each model separately: all six agree on the direction of every substantive effect, with civic commitment increasing between +9.3 and +18.1 percentage points across all models and countries. Even the smallest model (Qwen 2.5 3B) produces treatment effect estimates in the correct direction for every code in every country. The majority-vote procedure is thus a conservative safeguard, not a correction for disagreement. Results are stable across alternative thresholds (3/6, 5/6) and when GPT-4o-mini is excluded entirely (Supplementary Information, Appendix E). A simple keyword/dictionary baseline detects the same treatment direction but attenuates the magnitude

by half ($\kappa = 0.51$ with the LLM majority vote), which shows that LLMs outperform keyword matching on this task.

Triangulation

The two approaches answer different questions. STM shows the empirical structure of the text without imposing prior categories. It can surface patterns the researcher did not anticipate. But STM cannot tell you whether a topic corresponds to a specific theoretical construct, and on very short text (median 6–10 characters), topic assignments are often diffuse. LLM classification maps responses onto theory-derived categories with greater precision, but only if the categories are right. An ill-specified codebook produces precise but misleading estimates.

Running both methods on the same data is useful precisely when they disagree. The STM results in this study showed that language and residence topics move in the *opposite* direction from civic engagement topics under the treatment. That empirical pattern, which emerged without any prior categorization, made clear that the v1 codebook was grouping together constructs that respond differently to the experimental manipulation. LLM classification alone would not have flagged this problem, because the model faithfully follows whatever codebook it is given. The STM provided the evidence needed to refine the measurement before drawing conclusions from it.

This is not triangulation in the sense of independent data sources confirming the same finding; both methods operate on the same text corpus. It is a workflow in which each method has a specific role: STM identifies the empirical structure, LLM classification applies theoretical categories informed by that structure, and the combination catches errors that either method alone would miss. The key difference from single-stage LLM classification is the feedback loop: the unsupervised step diagnoses problems in the codebook that the supervised step, on its own, would faithfully reproduce.

A structural difference between the two methods also explains why they produce different effect magnitudes. STM assigns every document a probability distribution over all topics, including uninformative responses. Noise gets distributed across the topic space rather than isolated, producing conservative treatment effect estimates that reflect the full corpus (Roberts et al., 2014). LLM classification includes an

explicit `other` category that filters uninformative responses out of the substantive codes, so treatment effects are measured against a cleaner denominator. The full-sample STM estimates are not wrong; they reflect a different estimand. The Supplementary Information (Appendix B) provides a sensitivity analysis showing that as diffusely assigned STM documents are progressively excluded, the STM civic effect increases monotonically and converges toward the LLM estimate.

Results

STM: What Did Respondents Write About?

The South Korean model ($K = 6$) identifies six topics that map onto conjoint profile attributes and respondent evaluation criteria. The largest topics concern identity and practical credentials: religious/ethnic preferences (22.9%), residence and assimilation effort (21.3%), and community engagement and volunteering (18.8%). Employment and family stability (13.1%), professional experience (12.1%), and language/visa considerations (11.9%) account for the rest. Taiwan's model ($K = 8$) produces a comparable structure, with language fluency and first impressions (18.6%), work experience and social integration (18.5%), and civic activities and ethnicity (14.6%) as the largest topics.

That respondents wrote about the attributes they saw in the immigrant profiles is itself a finding. It means they processed the experimental stimuli and evaluated candidates on the dimensions the conjoint design intended to test.

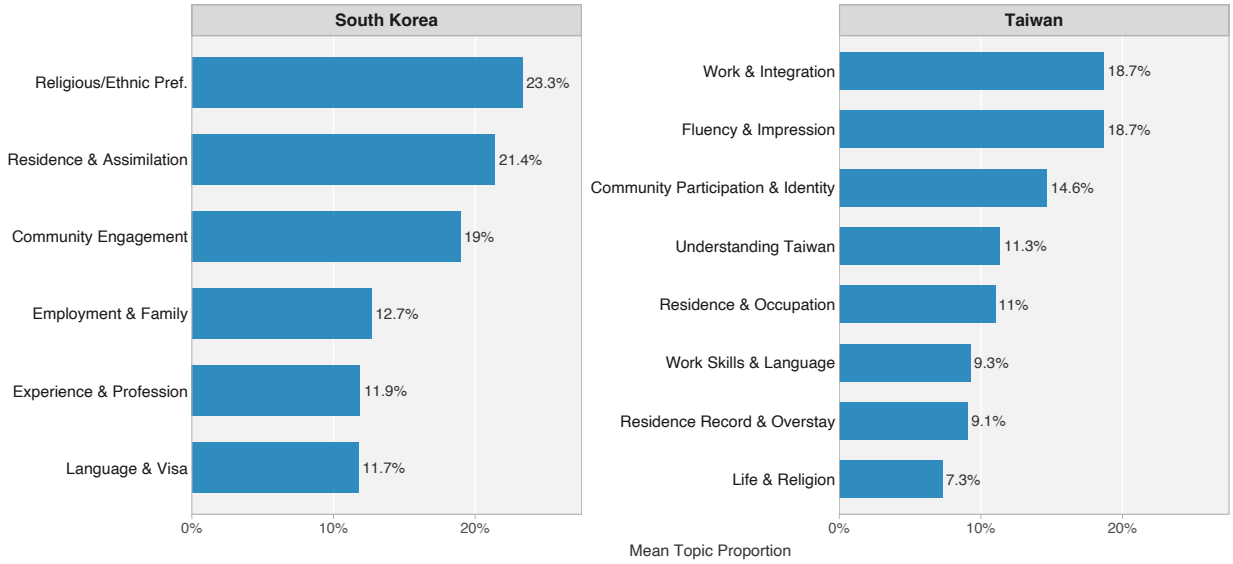

Figure 1: STM topic prevalence by country. Mean document-topic proportions for South Korea ($K = 6$) and Taiwan ($K = 8$).

STM: Effect of the Social Integration Cue

In South Korea, the social integration cue increases discussion of Community Engagement by 6.7 percentage points (pp) while decreasing discussion of Language & Visa (-4.0 pp) and Employment & Family (-2.3 pp). All three effects are statistically significant.

Taiwan shows the same pattern at greater magnitude. The cue increases Community Participation & Identity by 11.4 pp, Work & Integration by 3.1 pp, and Understanding Taiwan by 1.8 pp, while decreasing Fluency & Impression (-5.4 pp), Residence & Occupation (-4.7 pp), Residence Record (-3.0 pp), and Work Skills (-2.7 pp). All effects are significant at the 95% level; see Figure 2.

The pattern is consistent across both countries, and the *substitution* effects are as informative as the civic increase. If the open-text responses merely reflected attribute recall, respondents parroting back what they saw, one would expect the civic cue to add civic language without displacing other themes. Instead, language, residence, and employment topics all decline, suggesting that the cue reshapes the evaluative framework rather than adding a talking point. This substitution pattern is the strongest evidence that the social integration attribute affects how respondents reason about the profiles, not just what they mention afterward.

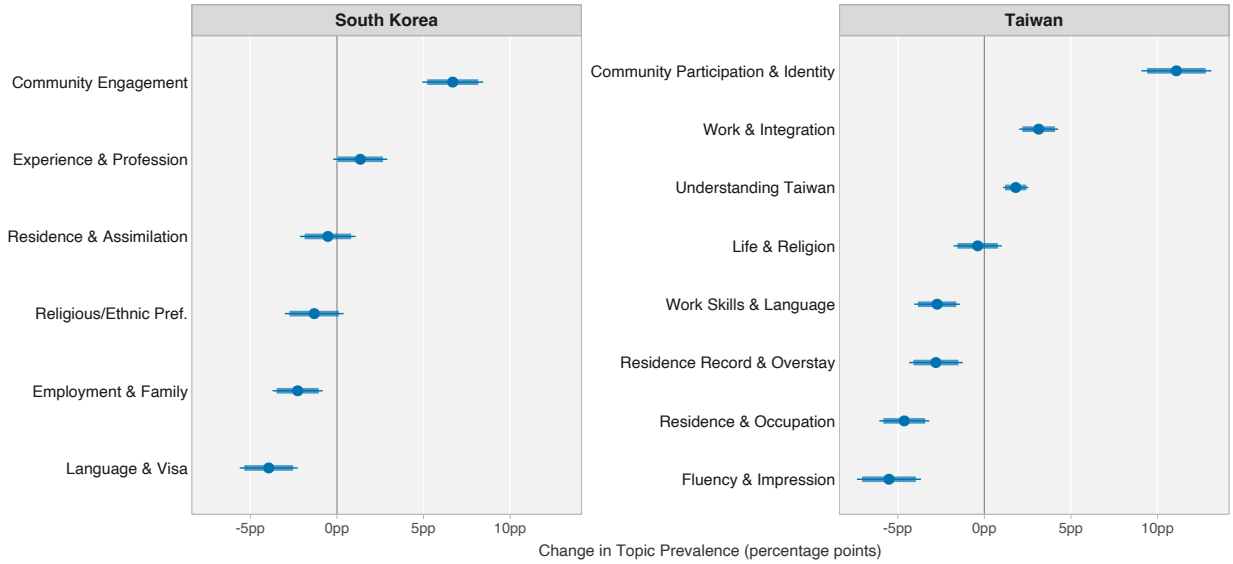

Figure 2: Effect of social integration cue on topic prevalence. STM coefficient estimates with 90% (thick) and 95% (thin) confidence intervals.

LLM Classification: Code Prevalence

Using majority-vote classification (4+ of 6 models agree), functional integration is the most common substantive code in both countries (∼32%), followed by identity concern (15–21%), economic focus (19–23%), and other (20–21%). Civic commitment (active behavioral engagement) accounts for 2.0% (South Korea) and 1.0% (Taiwan) when the cue is *not* shown, rising to 16.6% and 19.2% when it *is* shown.

Separating functional integration from civic commitment makes clear that practical attributes (language, residence, law-abiding behavior) dominate respondent reasoning in both countries, distinct from the active civic engagement that the cue is designed to activate.

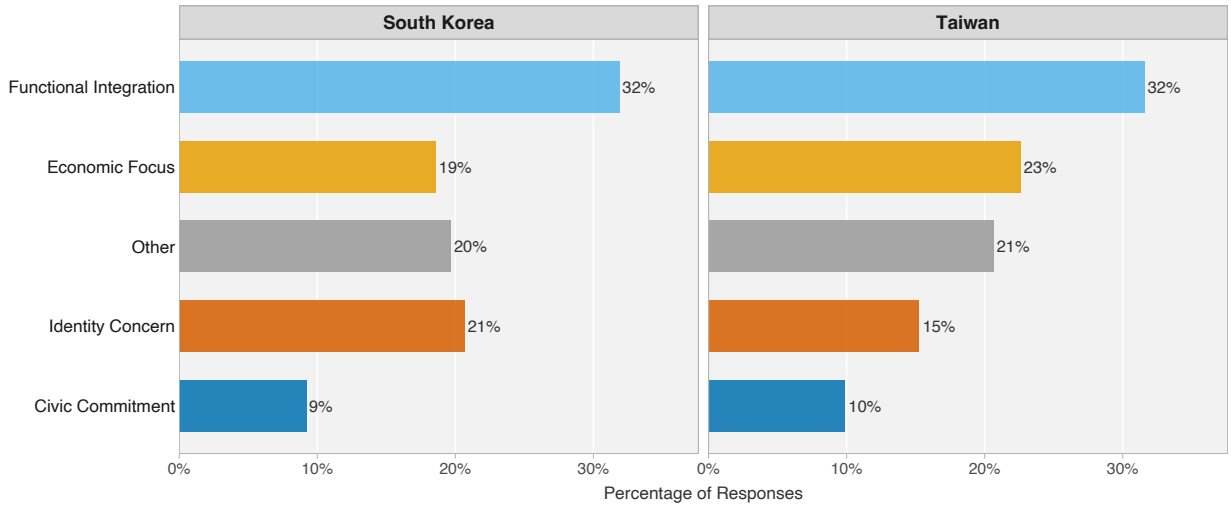

Figure 3: LLM classification: code prevalence by country. Majority-vote classification (4+ of 6 models agree).

LLM Classification: Effect of the Social Integration Cue

In South Korea, the cue increases civic commitment by +14.7 percentage points (from 2.0% to 16.6%), while decreasing functional integration by -7.9 pp and economic focus by -5.6 pp. Identity concern decreases modestly (-2.7 pp).

The Taiwanese results are similar in direction but stronger in magnitude: civic commitment increases by +18.2 pp (from 1.0% to 19.2%), while economic focus decreases by -11.4 pp and functional integration by -6.8 pp. Identity concern is stable (-1.0 pp).

All prevalence figures and treatment effects include the `other` category in the denominator. Table C.7 in the Supplementary Information reports the same effects computed among substantive codes only (excluding `other`), which produces larger point estimates.

Bootstrap confidence intervals place all of these effects outside zero at the 95% level. Across the six individual models, civic commitment effects range from +9.3 to +16.1 pp in South Korea and +14.4 to +18.1 pp in Taiwan, all significant. The treatment effect is present across all response lengths, including responses of five characters or fewer ($+7$ pp), though it is larger for longer responses (Supplementary Information, Appendix E).

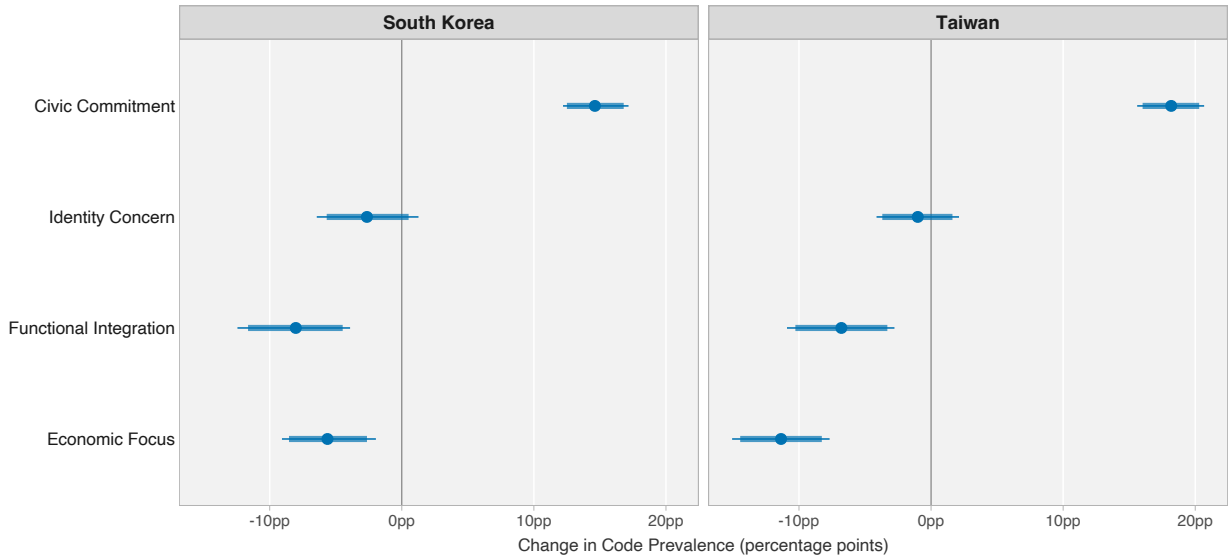

Figure 4: Effect of social integration cue on LLM-classified response themes. Majority-vote classification (4+ of 6 models). Bootstrap 90%/95% CIs (2,000 iterations).

All six LLMs produce consistent cue effect estimates (Figure 5; see also Table C.6 in the Supplementary Information for the full per-model breakdown). All six models agree that the social integration cue significantly increases civic commitment and decreases economic focus in both countries. Pairwise agreement ranges from 66.3% to 91.2% (Cohen's κ: 0.578–0.886), with Fleiss' multi-rater $\kappa = 0.745$ indicating substantial agreement.

Agreement increases with model size. Within the Qwen 2.5 family, pairwise κ with GPT-4o-mini rises monotonically: 0.744 (3B), 0.841 (32B), 0.860 (72B). The highest agreement overall is between Qwen 2.5 32B and 72B ($\kappa = 0.886$, 91.2% raw agreement), and the strongest cross-family pair is GPT-4o-mini and Qwen 2.5 72B ($\kappa = 0.860$, 89.1%). Larger open-weight models converge toward each other and toward the proprietary model. Chae and Davidson (2025) report a similar pattern.

Overall, 92.8% of responses receive a majority classification (4+ of 6 models agree). The weakest performer is Llama 3.1 8B, which inflates functional integration classifications in Taiwan (43.7% vs. 30–37% for the four larger models), probably due to weaker CJK language capabilities. Yet even this model produces civic commitment effects in the correct direction (+12.2 pp in South Korea, +15.9 pp in Taiwan), and the smallest model in the study (Qwen 2.5 3B) agrees on the direction of every effect in every country. For short, semantically straightforward open-text responses, model size matters less for directional

conclusions than for precise agreement on individual classifications.

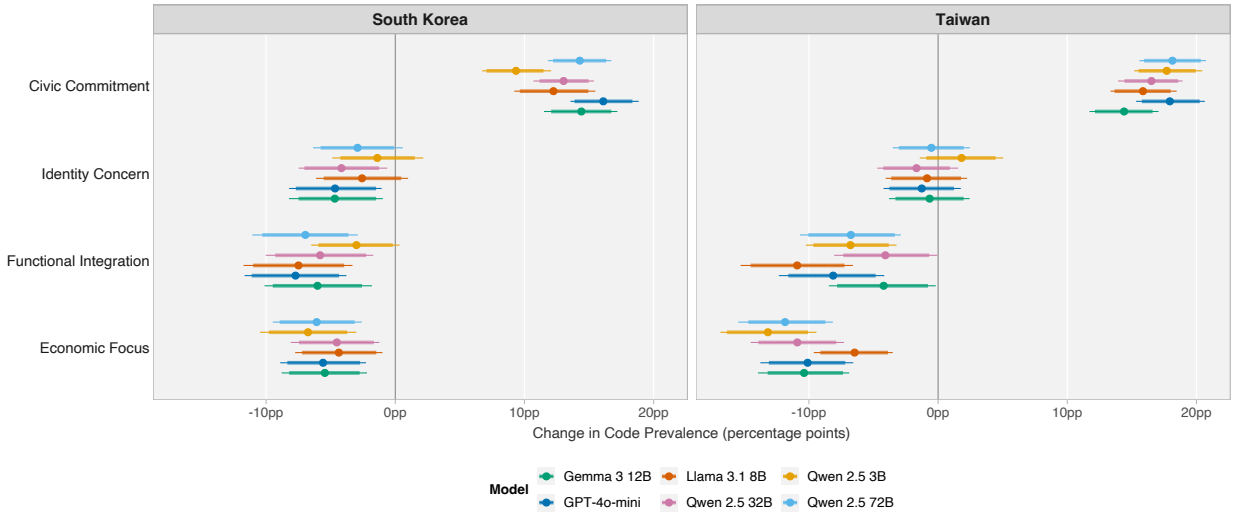

Figure 5: Cross-model robustness: cue effect estimates from six diverse LLMs. All models produce consistent treatment effect estimates for civic commitment.

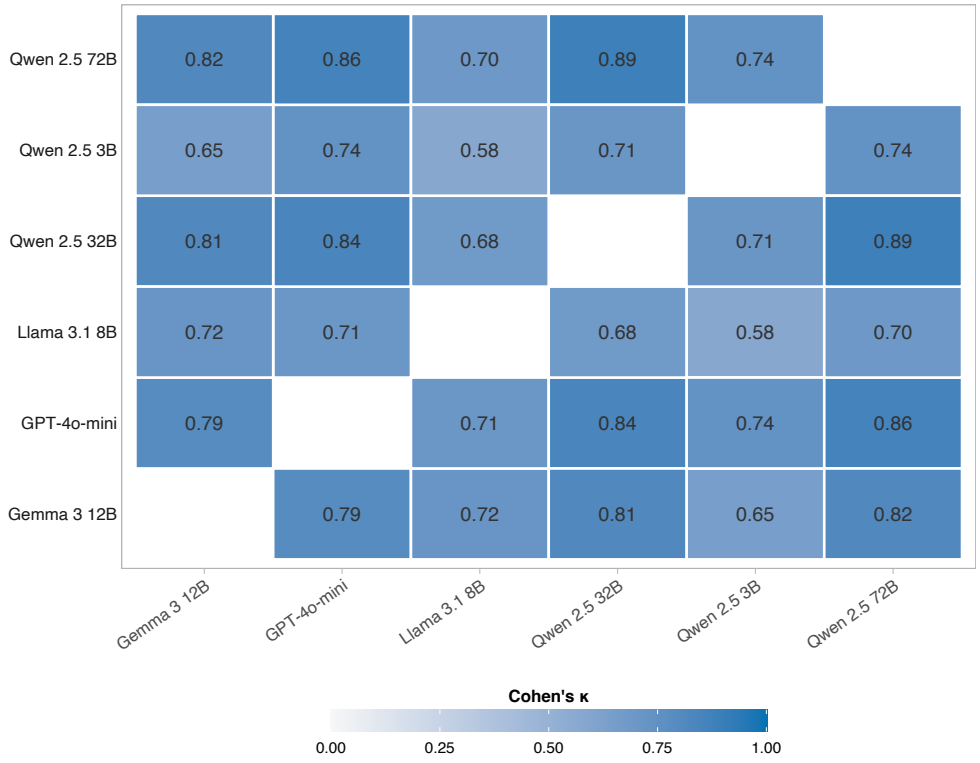

Figure 6: Cross-model agreement: pairwise Cohen's κ for six LLMs. Fleiss' multi-rater $\kappa = 0.745$.

Triangulation

The cleanest comparison between methods is for civic engagement, where a single STM topic maps directly to the LLM `civic_commitment` code in each country. In South Korea, the STM "Community Engagement" topic increases by 6.7 pp while the LLM civic code increases by 14.7 pp. In Taiwan, the STM "Community Participation & Identity" topic increases by 11.4 pp while the LLM civic code increases by 18.2 pp. All effects are significant. Both methods agree on the direction, significance, and cross-country pattern: the cue activates civic reasoning, and the effect is larger in Taiwan.

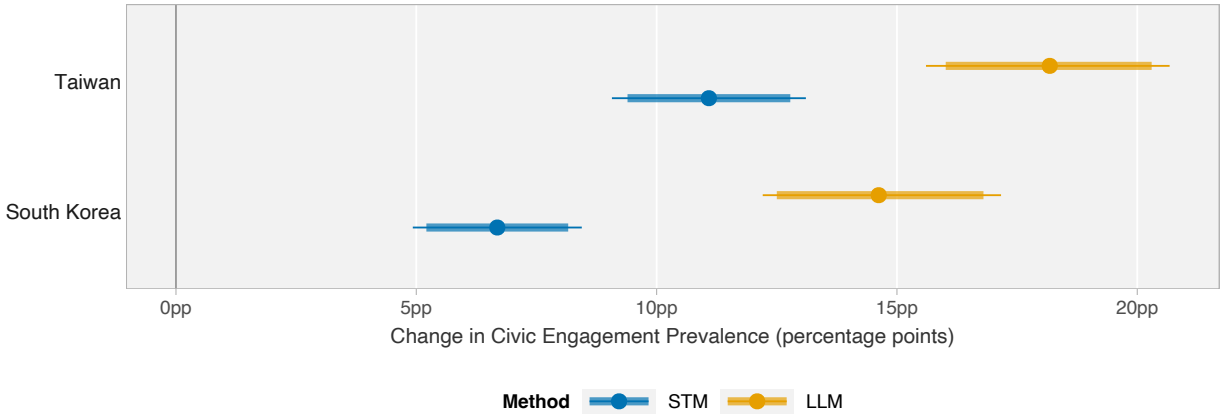

Figure 7: Triangulation: STM civic topic and LLM civic commitment treatment effects. The comparison where the topic-to-code mapping is unambiguous. Thick bars: 90% CIs; thin bars: 95% CIs.

Two factors explain why LLM estimates are larger than STM estimates. First, LLM codes are broader than individual STM topics: `civic_commitment` captures all civic reasoning in a single code, while STM distributes related content across multiple topics. Second, STM assigns topic proportions to every document regardless of content quality, while LLM classification routes uninformative responses to `other`, producing cleaner substantive codes and sharper treatment effects. This can be measured directly: the maximum topic probability ($\theta_{\max}$) for each document provides a measure of assignment clarity, and the median $\theta_{\max}$ is only 0.315 in South Korea and 0.303 in Taiwan (Supplementary Information, Appendix B). When diffusely assigned documents are progressively excluded, the STM civic effect rises monotonically and crosses the LLM estimate at $\theta_{\max} \approx 0.35$–$0.40$, which confirms that the gap between the two methods reflects how each handles uninformative text, not substantive disagreement about the

treatment effect.

For the remaining categories, the mapping between STM topics and LLM codes is not one-to-one, but the directional agreement is clear. On the practical and economic side, STM topics related to language, employment, and residence all decline when the cue is shown; the LLM similarly shows functional integration and economic focus declining. Identity concern is stable in both methods. Effects are consistently larger in Taiwan.

The v2 codebook refinement sharpens this convergence. By separating functional integration from civic commitment, the LLM classification now mirrors the STM finding that language and residence topics *decrease* when the cue is shown, which confirms that these practical attributes behave as substitutes for, not signals of, civic engagement.

Portability: A Standard Conjoint without Randomized Attributes

The primary analysis exploits within-attribute treatment randomization, a feature that most published conjoint experiments do not share. A validation framework that required such randomization would therefore apply only to a narrow subset of existing designs, and the methodological contribution would shrink accordingly. To test whether the approach transfers to the more common setup, I apply it to an earlier conjoint on immigrant *admission* in South Korea (Denney and Green, 2021). Respondents evaluated paired profiles on seven attributes (reason for application, country of origin, language skills, profession, employment plans, gender, and ethnicity) and explained their choices in an open-text response. All attributes are always shown, so there is no within-attribute cue to exploit. The question becomes whether the two-stage workflow can recover the reasoning structure of a standard conjoint and whether it can refine the categories an earlier analysis used to describe that reasoning.

I classify the open-text responses using the same six LLM classifiers with a five-code codebook adapted to the admission context: sociotropic (social harmony, community benefit), economic (employment, fiscal concerns), integration (language, cultural adaptation), identity (ethnicity, national origin, co-ethnic preference), and other. Cross-model agreement is high ($\kappa = 0.737$ for GPT-4o-mini vs. Qwen 2.5 3B).

LDA topic modeling ($K = 6$) on the Korean text produces topics that converge with the LLM codes: two topics map to functional integration (36% of documents), one to identity (17%), one to sociotropic reasoning (16%), one to economic concerns (13%), and one is mixed.

Table 4 shows the code distributions from five classifiers. Integration (18–27%), identity (20–22%), and economic reasoning (13–25%) each account for substantial shares. The "other" category captures vague or non-substantive responses (17–27%).

Code	**GPT-4o-mini**	**Qwen 3B**	**Gemma 12B**	**Qwen 32B**	**Qwen 72B**
Sociotropic	11.0	14.8	20.4	14.9	12.9
Economic	20.1	12.5	22.1	25.1	19.3
Integration	19.5	27.3	18.7	18.4	20.7
Identity	22.3	21.6	21.8	21.9	19.7
Other	27.0	23.8	17.0	18.8	27.3

Table 4: Code prevalence (%) for the immigrant admission conjoint ($N = 1{,}008$). What Denney & Green (2021) grouped as "sociotropic" reasoning here separates into sociotropic, economic, and integration codes.

The result sharpens the original study's findings. Denney and Green (2021) described a broad category of "sociotropic" reasoning without distinguishing between social-harmony arguments, economic arguments, and integration-effort arguments. Applying the five-code framework retrospectively decomposes this category into three distinct constructs. This is construct validation in a second sense: not only testing whether an experimental cue shifts reasoning as theorized, but asking whether the measurement categories used in a completed study capture the reasoning the data actually contain. The framework is therefore not limited to designs with treatment randomization. It can also be applied to existing conjoint data to refine constructs that were too coarse in the original analysis, which extends its methodological reach well beyond the subset of designs with within-attribute randomization. Full results are in Supplementary Information, Appendix D.

Calibration: LLM Classifications against Expert Human Coding

Cross-model agreement is a useful substitute when expert human coding is unavailable, but it is not a direct test of accuracy. Six models may agree because they have learned the same patterns from overlapping pretraining corpora, not because their classifications match what a careful human expert would produce. To test how LLM classifications perform against an independent human benchmark, and to check whether the iterative codebook workflow developed in the main analysis generalizes to a different dataset, I apply the framework to a conjoint experiment on South Korean attitudes toward North Korean migrant entrepreneurship policy (Denney et al., 2024). After the conjoint task, respondents answered an open-text question explaining their policy choice. Two human coders independently classified all responses into four categories: economic rationale, normative rationale, other rationale, and uninformative. Their inter-coder agreement was high (Cohen's $\kappa = 0.881$, 92.3% raw agreement), which provides a reliable benchmark.

I classified the same responses using GPT-4o-mini and Llama 3.1 8B with a four-code codebook matching the human coding scheme. The LLMs classified English translations of the responses, matching the input the human coders used. I then compared LLM classifications to the human consensus. Because this validation uses English text, it confirms that the codebook and classification logic are well-calibrated but does not directly validate the multilingual classification pipeline that produces the main results.

The codebook required iteration. The initial version (v1) produced only fair agreement with the human coders (GPT-4o-mini $\kappa = 0.40$; Llama 3.1 8B $\kappa = 0.52$). Examining disagreements showed that both LLMs interpreted "economic" too narrowly and "uninformative" too broadly relative to the human coding. Because the conjoint is entirely about economic policy, the human coders classified policy-effectiveness language (e.g., "this policy would help them succeed") as economic, while the LLMs treated it as normative or uninformative. Broadening the economic category to include policy-effectiveness reasoning and tightening the uninformative code to cover only genuinely empty responses closed the gap across two rounds of revision.

Comparison	Cohen's κ	Raw agreement	Landis & Koch
Human–Human	0.881	92.3%	Almost perfect
GPT-4o-mini–Human	0.673	78.8%	Substantial
Llama 3.1 8B–Human	0.611	74.5%	Substantial
GPT-4o-mini–Llama 3.1 8B	0.717	81.5%	Substantial

Table 5: Inter-rater agreement for the N.K. entrepreneurship conjoint (v3 codebook, after two rounds of codebook revision). Human–human agreement from Denney et al. (2024). Note: N.K. = North Korea.

After calibration, both LLMs fall in the "substantial" agreement range ($\kappa > 0.6$), and the two LLMs agree with each other at $\kappa = 0.717$. A gap remains between LLM-human and human-human agreement, but the trajectory across codebook versions is the more important finding: two rounds of refinement moved GPT-4o-mini from $\kappa = 0.40$ (fair) to $\kappa = 0.67$ (substantial), a pattern that parallels the v1-to-v2 improvement in the main analysis. The codebook, not the model, is the primary determinant of classification quality. Table 6 shows the code distributions from four classifiers. Economic rationale dominates (50–60%), consistent with the conjoint being entirely about economic policy. Normative rationale accounts for 16–22%, and the uninformative category is small (2–7%), confirming that the codebook revision successfully tightened this residual category.

Code	Llama 8B	Gemma 12B	Qwen 32B	Qwen 72B
Economic rationale	49.8	60.0	55.4	52.3
Normative rationale	22.3	21.6	18.5	16.0
Other rationale	22.5	16.4	19.3	26.7
Uninformative	5.3	1.9	6.8	5.0

Table 6: Code prevalence (%) for the N.K. entrepreneurship conjoint ($N = 2{,}009$). Economic rationale dominates, consistent with the policy domain. Note: N.K. = North Korea.

This result has two implications for the main analysis. First, it provides external evidence that the iterative codebook workflow advocated in this paper produces meaningful gains in classification accuracy.

The $0.40 \rightarrow 0.67$ trajectory observed here, moving from fair to substantial agreement, parallels the v1-to-v2 refinement in the naturalization analysis. In both cases the decisive improvement came from rewriting the codebook, not from switching to a larger or more capable model: the codebook, not the model, is the primary lever for improving classification quality, which is the load-bearing methodological claim of the paper. Second, it calibrates expectations about absolute performance. Even after careful iteration, LLM-human agreement ($\kappa \approx 0.6$–0.7) remains below expert human-human agreement ($\kappa \approx 0.9$). The six-model cross-validation in the main analysis, which achieves Fleiss' $\kappa = 0.745$ across models of very different size and architecture, is consistent with this range and suggests that multi-model consensus compensates for the gap between any single LLM and a human expert.

The NK dataset also provides the strongest test of whether token-level logprobs track classification uncertainty. Against the human consensus, all six models are overconfident: they assign high predicted probabilities to classifications that disagree with human coders at substantial rates (Expected Calibration Error 0.15–0.26; Supplementary Information, Table **??**). Larger models are better calibrated (Qwen 2.5 72B ECE = 0.145 vs. Qwen 2.5 3B ECE = 0.260), but none are well-calibrated in absolute terms. By contrast, calibration against the majority vote on the SK/TW corpus produces much lower ECE (0.01–0.10), reflecting shared uncertainty patterns across models with overlapping pretraining data. The gap between these two benchmarks confirms that inter-model agreement overstates classification reliability, and that logprobs, while informative about relative difficulty, should not be interpreted as true probabilities of correctness.

Discussion and Conclusion

Open-weight models produce results comparable to GPT-4o-mini, with all models agreeing on the direction and significance of treatment effects. Researchers need not depend on proprietary APIs, whose outputs may shift between model versions (Barrie et al., 2025), to conduct rigorous text classification. The open-weight models are the reproducible core; GPT-4o-mini confirms that these alternatives match the proprietary models used in current practice.

Model size matters for agreement, not for direction. Within the Qwen 2.5 family, pairwise κ with GPT-4o-mini rises monotonically from 0.744 (3B) to 0.860 (72B), consistent with Chae and Davidson (2025). Yet every model, including the 3B one runnable on a consumer GPU, produces civic commitment effects in the correct direction for every code in every country. Small models suffice when the goal is directional treatment effects on short open-text responses; mid-range models (12B–32B) offer a reasonable tradeoff for more fine-grained applications.

Multi-model majority voting also offers a practical alternative to human-coded ground truth. When six models of different sizes, architectures, and training data agree on a classification, as they do for 93% of responses here, that consensus provides usable measurement without expert coders. The ground-truth extension confirms the gap between multi-model consensus and expert human coding is modest ($\kappa \approx 0.6$–0.7) and that codebook refinement closes much of it.

This approach complements existing tools for assessing respondent engagement. Prior work validates conjoint experiments against real-world behavior (Hainmueller et al., 2015) and examines visual attention using eye-tracking (Jenke et al., 2021). Open-text responses provide a different window: how respondents *reason* about profile attributes, not just whether they attended to them. Unlike manipulation checks that condition on post-treatment responses and risk selection bias (Aronow et al., 2019), open-text validation examines the full sample. Other work scales open-ended responses on a continuous dimension (DiGiuseppe and Flynn, 2026); the approach here is categorical, aimed at what kind of reasoning respondents use and whether the treatment shifts it.

The iterative codebook design process makes a broader point. Running an unsupervised model first can catch coding mistakes that would otherwise go unnoticed. The v1 codebook grouped civic engagement with language and residence under a single code; the STM showed that these move in opposite directions under the treatment, so the v2 codebook separated them. That single revision changed more classification outcomes than any other design choice in the analysis.

Several limitations bear noting. Post-choice open-text elicitation invites post-hoc justification rather than real-time introspection; the framework establishes that treatment shifts what respondents *write*, which is necessary but not sufficient evidence that it shifts what they *think*. The substitution pattern,

where civic cues displace language, residence, and economic reasoning rather than merely adding civic language, is harder to explain as mere recall. Responses are short (median 6–10 characters) and self-generated, so STM assignments are often diffuse and some considered attributes may go unwritten; the θ_{max} sensitivity analysis in SI Appendix B quantifies the effect on STM estimates. Individual model performance on CJK text varies, and majority-vote aggregation mitigates but does not eliminate model-specific biases. Token-level log-probabilities track inter-rater difficulty ($r_s = 0.46$) but are overconfident against human ground truth (ECE 0.15–0.26), so they should not be taken as true probabilities of correctness. Scope is also limited: the framework has been tested on immigration conjoints in two East Asian democracies with short, concrete responses; transferability to more abstract constructs, discursive responses, or lower-resource languages is open.

A final concern is response authenticity in the agentic AI era. The surveys analyzed here pre-date widespread access to agentic tools, so AI-generated contamination is not a threat to the present findings, but Westwood (2025) show that agentic systems now complete surveys while passing 99.8% of standard attention checks. A related concern is delegation: respondents who outsource open-text answers to a language model rather than writing their own.[3] If respondents delegate their reasoning to an LLM, the analyst ends up using LLMs to classify LLM-generated text. Researchers adopting this framework should treat authenticity as an assumption to be actively verified, paired ideally with behavioral signals or design choices that make genuine participation less effortful than delegation.

Combining structural topic modeling with LLM classification, and using each to inform the other, lets researchers test whether the construct an experiment is designed to measure is the construct respondents reason about, and whether the design activates that reasoning rather than something adjacent. Both methods converge in the primary application: the social integration cue shifts respondent reasoning from practical and economic criteria toward active civic engagement. The ground-truth validation confirms that iterative codebook refinement produces substantial LLM-human agreement; the codebook, not the model, is the primary lever for improving classification quality. Token-level log-probabilities reveal which

[3] The distinction between deception and delegation as separate threats to survey research in the agentic era is developed by Yamil Velez, "Who Answered This Survey? Deception, Satisficing, and Delegation in the Agentic Era," *New Instruments* (Substack), March 30, 2026, `https://newinstruments.substack.com/p/who-answered-this-survey`.

classifications are uncertain, information that self-reported confidence labels do not reliably convey. Researchers should consider embedding open-text questions in their survey experiments; the smallest open-weight model tested here runs on a consumer GPU and produces directionally consistent results, and the validity evidence is worth the modest additional burden.

Data Availability

Replication data, code, codebooks, and all analytical outputs are available in the replication archive at [URL/DOI to be added upon acceptance]. The archive includes the analysis-ready corpus, all LLM classification outputs, STM intermediate files, and publication figures.

Ethics Statement

This study is a secondary analysis of pre-existing survey data collected for other research projects (Denney et al., 2025; Denney and Green, 2021; Denney et al., 2024). No new human subjects data were collected. The original surveys were conducted in accordance with applicable ethical standards and received appropriate institutional approvals.

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Competing Interests

The author declares none.

Use of AI-Assisted Tools

Claude Code (Anthropic, Opus 4.6) was used during the preparation of this manuscript in two capacities: to assist with code cleaning and formatting for the replication archive, and for analysis confirmation

and checking and for technical and substantive edits to improve the precision of the manuscript's language. All substantive ideas, analytical decisions, research design, and core writing are the author's own. The author reviewed and revised all AI-assisted output and takes full responsibility for the content of this publication.

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Supplementary Information

"What Were They Thinking?" Using Open-Text Responses to Validate Constructs in Survey Experiments

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17 April 2026

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A Survey Design and Sample

Respondents were recruited via Qualtrics using stratified quota sampling by sex, age, and geographic region to approximate national census distributions. Table A.1 presents the sample characteristics for both countries.

	South Korea ($N = 1{,}999$)	**Taiwan** ($N = 2{,}050$)
Gender		
Male	1,015 (50.8%)	1,094 (53.4%)
Female	984 (49.2%)	956 (46.6%)
Age		
Mean	46.6	41.3
Range	19–84	14–88
Education		
College or higher	1,610 (80.6%)	1,506 (73.5%)
Below college	387 (19.4%)	544 (26.5%)
Region		
Capital area / Northern TW	1,484 (74.3%)	1,082 (52.8%)
Other regions	513 (25.7%)	968 (47.2%)
Treatment assignment		
Cue shown	996 (49.8%)	1,007 (49.1%)
Cue not shown	1,003 (50.2%)	1,043 (50.9%)

Table A.1: Sample characteristics. Age is in years at time of survey (2024). "Capital area" for South Korea includes Seoul, Gyeonggi, and Incheon. "Northern TW" includes Taipei, New Taipei, and Taoyuan.

Table A.2 presents the full set of 12 attributes used in the conjoint experiment on immigrant naturalization, with all attribute levels. Profiles were shown to respondents in South Korea and Taiwan; respondents chose which immigrant they considered "more deserving of citizenship." All attributes and levels were independently randomized across profiles.

Attribute	Levels
Social integration	Only associates with fellow expatriates; Often spends time alone; Co-hosts monthly multicultural events; Volunteers weekly at a local community center
Ethnicity	Co-ethnic; Not co-ethnic
Religion	Not religious; Christian; Buddhist; Muslim
Country of origin	United States; China; Indonesia; Brazil (SK) / South Korea (TW); the Philippines
Occupation	Computer programmer; Farmer; Unemployed; Office worker
Family status	Single; Married; Married with children
Age	27; 36; 47; 58
Sex	Male; Female
Prior work experience	None; Worked in home country; Worked previously in country
Prior trips	Visited once with friends; Previously overstayed visa; Visited on short-term working visa
Years since arrival	2; 7; 12
Language ability	None; Broken; Conversational; Fluent

Table A.2: Conjoint experiment: full attribute table (12 attributes with levels). The social integration attribute was randomly withheld from some profiles, creating the cue-shown vs. cue-not-shown treatment comparison. Country of origin levels differ slightly between the South Korea and Taiwan surveys.

B Structural Topic Models

B.1 Topic labels and top terms

Tables B.1 and B.2 present the topic labels and mean prevalence for each STM model. Top terms (FREX and probability) are in Korean and Traditional Chinese, respectively, and are available in the replication code.

Topic	Label	Prevalence (%)
1	Community Engagement	18.8
2	Language & Visa	11.9
3	Religious/Ethnic Preference	22.9
4	Residence & Assimilation	21.3
5	Experience & Profession	12.1
6	Employment & Family	13.1

Table B.1: South Korea STM topics ($K = 6$). Prevalence is the mean document-topic proportion across the corpus.

Topic	Label	Prevalence (%)
1	Fluency & Impression	18.6
2	Life & Religion	6.2
3	Work & Integration	18.5
4	Residence & Occupation	9.7
5	Understanding Taiwan	11.4
6	Work Skills & Language	8.1
7	Community Participation & Identity	14.6
8	Residence Record & Overstay	12.8

Table B.2: Taiwan STM topics ($K = 8$). Prevalence is the mean document-topic proportion across the corpus.

B.2 Topic assignment clarity and threshold sensitivity

Structural topic models assign every document a probability distribution over all K topics, regardless of content quality. A substantive response that clearly discusses a single theme receives a concentrated

distribution (high $\theta_{\max}$), while an uninformative or ambiguous response receives a diffuse distribution spread across topics. This section examines the clarity of STM topic assignments and how treatment effect estimates change when diffusely assigned documents are excluded.

The open-text responses in this study are extremely short. After morphological analysis (kiwipiepy for Korean, quanteda for Chinese) and stopword removal, 41% of South Korean documents and 30% of Taiwanese documents contain a single token; the median is 2 tokens in both countries. Standard STM preprocessing drops documents that produce zero features (265 of 1,999 in South Korea, 145 of 2,050 in Taiwan), but no minimum word count threshold is applied, consistent with standard practice in the STM literature (Roberts et al., 2014). STM handles short documents through its Bayesian prior: documents with few words receive posterior topic proportions dominated by the prior rather than the data, producing appropriately diffuse assignments that express the model's uncertainty.

Figure B.1 shows the distribution of $\theta_{\max}$ (the maximum topic probability for each document) in both countries. The dashed orange line marks the uniform distribution ($1/K$: 0.167 for South Korea, 0.125 for Taiwan), and the dotted blue line marks a moderate threshold of $\theta_{\max} = 0.30$. In South Korea ($K = 6$), the median $\theta_{\max}$ is 0.315, and only 3.2% of documents exceed 0.50. In Taiwan ($K = 8$), the median is 0.303 with 11.5% above 0.50. The diffuse assignments are a direct consequence of the short document lengths: with 1–2 tokens, the posterior cannot move far from the prior.

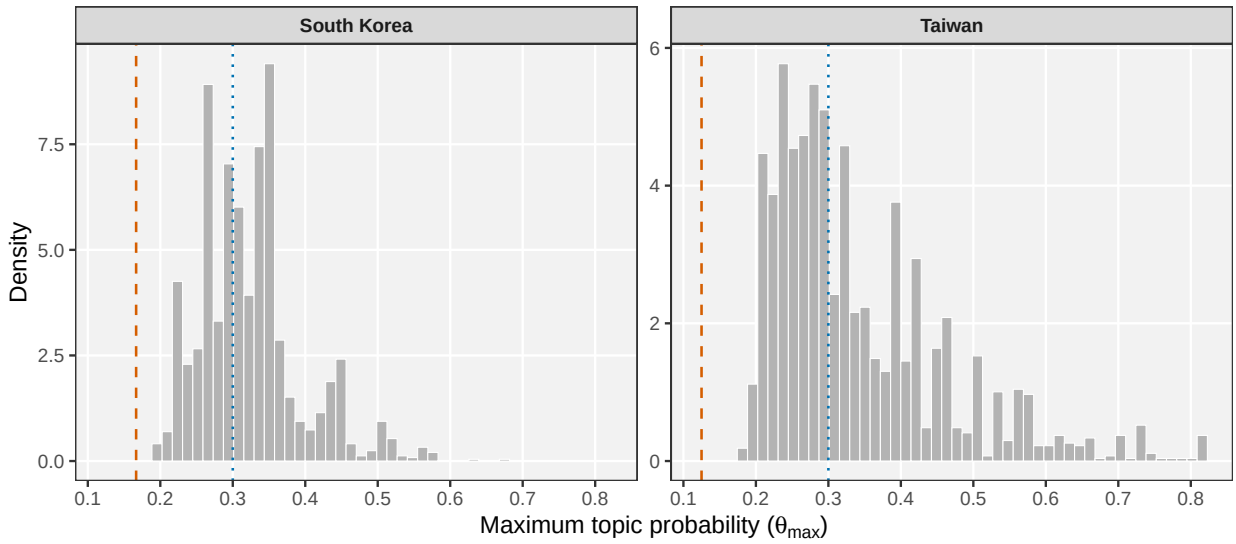

Figure B.1: Distribution of maximum topic probability ($\theta_{\max}$) for each document. Dashed orange: uniform baseline ($1/K$). Dotted blue: moderate filtering threshold ($\theta_{\max} = 0.30$). Most documents are diffusely assigned across topics, especially in South Korea where $K = 6$ constrains the topic space.

Figure B.2 shows how the civic topic treatment effect changes as the minimum $\theta_{\max}$ threshold increases (i.e., as diffusely assigned documents are progressively excluded). The solid lines show STM civic topic effects at each threshold; the dashed horizontal lines show the LLM majority-vote civic commitment estimates for comparison.

As the threshold rises, the STM civic effect increases monotonically in both countries. At $\theta_{\max} \geq 0.30$ (retaining 59% of South Korean and 51% of Taiwanese documents), the effects rise to +10.5 pp and +18.2 pp, respectively. The STM estimates cross the LLM estimates around $\theta_{\max} \approx 0.35$–$0.40$, and at higher thresholds the STM effects substantially exceed the LLM estimates (reflecting the smaller and potentially unrepresentative samples that remain).

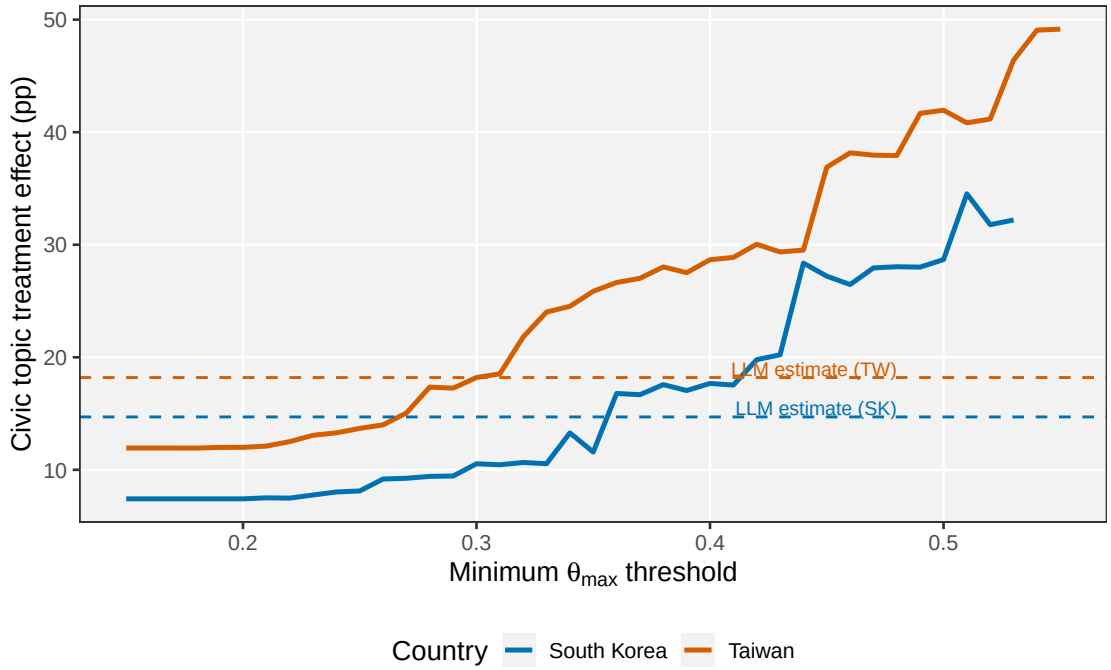

Figure B.2: Sensitivity of the civic topic treatment effect to $\theta_{\max}$ thresholds. Solid lines: STM civic topic effect (mean θ difference, cue-shown minus cue-not-shown). Dashed lines: LLM majority-vote civic commitment estimates. As diffusely assigned documents are excluded, the STM effect increases and approaches or exceeds the LLM estimate.

Table B.3 summarizes these results at selected thresholds.

Min. θ_{max}	**South Korea**		**Taiwan**	
	Docs retained	Civic effect (pp)	Docs retained	Civic effect (pp)
0 (all)	1,734 (100%)	$+7.4$	1,905 (100%)	$+11.9$
0.25	1,499 (86%)	$+8.1$	1,463 (77%)	$+13.7$
0.30	1,014 (59%)	$+10.5$	964 (51%)	$+18.2$
0.35	505 (29%)	$+11.6$	681 (36%)	$+25.9$
0.40	226 (13%)	$+17.7$	476 (25%)	$+28.7$
LLM majority-vote (for comparison)				
—	1,820 (91%)	$+14.7$	1,938 (94%)	$+18.2$

Table B.3: Civic topic treatment effect at different θ_{max} thresholds. STM effects are raw mean θ differences (cue-shown minus cue-not-shown), which differ slightly from the `estimateEffect()` regression coefficients reported in the main text (e.g., $+7.4$ here vs. $+6.7$ in the main text for South Korea at threshold 0). The STM effect increases as diffusely assigned documents are excluded. At $\theta_{\text{max}} \geq 0.30$, STM estimates approach the LLM majority-vote estimates. The LLM row shows the majority-vote civic commitment effect for comparison; the "docs retained" count for the LLM excludes responses without majority agreement (7.2% of the corpus).

This analysis illustrates a structural difference between the two methods. STM distributes every document, including uninformative ones, across the entire topic space. This dilutes treatment effects because noise documents receive non-trivial proportions on treatment-sensitive topics. LLM classification handles this differently: the explicit `other` category absorbs uninformative responses, effectively filtering them before treatment effects are computed on the substantive codes. The θ_{max} threshold analysis is the STM analogue of this filtering. As the threshold increases and noisy documents are removed, the STM estimates converge toward (and eventually exceed) the LLM estimates, confirming that the difference in effect magnitudes between the two methods is driven by how each handles uninformative text rather than by substantive disagreement about the treatment effect.

C LLM Classification

C.1 Codebook

The codebook follows the structured format recommended by Halterman & Keith (2025), with each code defined by a label, definition, clarification (inclusion criteria), and negative clarification (exclusion criteria). Each code also includes few-shot examples in both Korean and Traditional Chinese.

Code	Definition and Scope
`civic_commitment`	Participation in community life: volunteering, community organizing, social participation, cultural events, helping neighbors. The respondent cites engagement directed outward toward the community.
`functional_integration`	Individual-level adaptation: language competence, residence duration, law-abiding behavior. The respondent cites practical prerequisites for functioning in the host country, rather than participation in community life.
`identity_concern`	Ethnic, cultural, or national identity concerns. Includes religion and ethnicity when used as criteria for evaluating immigrants.
`economic_focus`	Economic contributions and qualifications: jobs, taxes, career, occupation, work experience, financial self-sufficiency.
`other`	Truly uninformative responses: filler words, gibberish, meta-comments about the survey. Brevity alone is never sufficient reason for this code.

Table C.1: LLM classification codebook (v2, 5 codes). The main refinement from v1 was separating `functional_integration` from `civic_commitment` based on STM evidence that language and residence topics decrease with the civic integration cue.

C.2 Infrastructure and parse errors

Table C.2 provides technical details for each classifier. All models used temperature $= 0$, seed $= 42$, and a maximum of 200 output tokens. Open-weight models were served via Ollama.

Model	Parameters	Infrastructure	Parse Errors
GPT-4o-mini	—	OpenAI API	0
Qwen 2.5 3B	3B	Consumer GPU (NVIDIA RTX 3060 Ti, 8GB)	38
Llama 3.1 8B	8B	Consumer GPU (NVIDIA RTX 3060 Ti, 8GB)	0
Gemma 3 12B	12B	University HPC, NVIDIA A100	1
Qwen 2.5 32B	32B	University HPC, NVIDIA A100, 4-bit quantization	0
Qwen 2.5 72B	72B	University HPC, NVIDIA A100 80GB, 4-bit quantization	0

Table C.2: LLM infrastructure and parse error counts. Parse errors indicate responses where the model output could not be parsed into a valid JSON classification; these responses are excluded from agreement and prevalence calculations. Qwen 2.5 3B accounts for nearly all parse errors (38 of 39 total across all models), consistent with smaller models producing less reliable structured output.

C.3 Per-model code distributions

Table C.3 presents the code distribution for each of the six LLM classifiers, by country.

Model	Country	Civic	Functional	Identity	Economic	Other
GPT-4o-mini	South Korea	11.7	28.5	21.2	17.5	21.0
GPT-4o-mini	Taiwan	10.8	31.1	14.6	21.1	22.3
Qwen 2.5 3B	South Korea	10.7	19.5	20.4	21.4	28.1
Qwen 2.5 3B	Taiwan	11.3	21.2	16.8	25.2	25.5
Llama 3.1 8B	South Korea	14.7	36.4	21.1	18.5	9.4
Llama 3.1 8B	Taiwan	10.7	43.7	16.2	14.8	14.6
Gemma 3 12B	South Korea	12.5	35.5	22.3	16.9	12.7
Gemma 3 12B	Taiwan	10.9	36.7	15.0	22.0	15.4
Qwen 2.5 32B	South Korea	7.5	33.7	20.0	18.7	20.1
Qwen 2.5 32B	Taiwan	9.0	32.1	15.6	22.8	20.5
Qwen 2.5 72B	South Korea	9.4	30.7	18.9	19.4	21.6
Qwen 2.5 72B	Taiwan	10.7	30.0	14.7	23.4	21.2

Table C.3: Code distribution by model and country (% of classified responses). Llama 3.1 8B shows elevated functional integration in Taiwan (43.7%) relative to the four larger models (30.0–36.7%), likely reflecting weaker CJK language performance. Qwen 2.5 3B shows elevated `other` rates (25–28%), consistent with lower classification confidence at smaller model sizes.

C.4 Cross-model agreement

Table C.4 presents pairwise raw agreement rates and Cohen's κ for all 15 model pairs.

Model 1	Model 2	Agreement (%)	Cohen's κ
GPT-4o-mini	Qwen 2.5 3B	79.7	0.744
GPT-4o-mini	Llama 3.1 8B	77.2	0.706
GPT-4o-mini	Gemma 3 12B	83.9	0.792
GPT-4o-mini	Qwen 2.5 32B	87.6	0.841
GPT-4o-mini	Qwen 2.5 72B	89.1	0.860
Qwen 2.5 3B	Llama 3.1 8B	66.3	0.578
Qwen 2.5 3B	Gemma 3 12B	71.8	0.646
Qwen 2.5 3B	Qwen 2.5 32B	77.2	0.710
Qwen 2.5 3B	Qwen 2.5 72B	79.6	0.741
Llama 3.1 8B	Gemma 3 12B	79.0	0.722
Llama 3.1 8B	Qwen 2.5 32B	75.7	0.683
Llama 3.1 8B	Qwen 2.5 72B	76.8	0.700
Gemma 3 12B	Qwen 2.5 32B	85.7	0.815
Gemma 3 12B	Qwen 2.5 72B	85.8	0.816
Qwen 2.5 32B	Qwen 2.5 72B	91.2	0.886
Fleiss' multi-rater κ			**0.745**

Table C.4: Pairwise agreement rates and Cohen's κ for all 15 model pairs ($N = 4{,}049$ responses). Agreement increases with model size: within the Qwen 2.5 family, κ with GPT-4o-mini rises from 0.744 (3B) to 0.841 (32B) to 0.860 (72B). The highest pairwise agreement is between Qwen 2.5 32B and 72B ($\kappa = 0.886$).

C.5 Model confidence: self-reported ratings

Each LLM classifier reports a self-assessed confidence level (HIGH, MEDIUM, or LOW) alongside its classification. Table C.5 summarizes the proportion of HIGH-confidence classifications by model and country.

Model	South Korea (% HIGH)	Taiwan (% HIGH)
GPT-4o-mini	80.9	87.9
Qwen 2.5 3B	69.0	69.9
Llama 3.1 8B	49.3	60.8
Gemma 3 12B	74.9	81.6
Qwen 2.5 32B	53.3	70.0
Qwen 2.5 72B	48.3	60.0

Table C.5: Proportion of HIGH-confidence classifications by model and country. GPT-4o-mini reports the highest confidence. Larger open-weight models do not uniformly report higher confidence than smaller ones, indicating that self-reported confidence reflects model-specific verbosity patterns rather than classification quality.

These self-reported confidence ratings are generated tokens, not probability estimates. The model produces what it judges to be a plausible human confidence label; this bears no necessary relationship to the model's actual predictive uncertainty. A model that reports HIGH confidence on 80% of responses is not necessarily more accurate than one that reports HIGH on 50%.

C.6 Model confidence: token-level probabilities and calibration

A better measure of model uncertainty comes from the token-level log-probabilities (logprobs) that language models assign during generation. When the model produces the classification label (e.g., `civic_commitment`) inside the JSON output, the log-probability of the first distinguishing token reflects the model's predictive distribution over the five candidate codes. Unlike the self-reported confidence rating, this probability is not a generated token but a readout of the model's softmax distribution at the point of classification.

For each of the six models and all three datasets, I extract the logprob of the first code token and convert it to a probability ($p = e^{\text{logprob}}$). The question is whether these probabilities are calibrated: does the predicted probability correspond to actual classification accuracy?

Calibration against inter-rater agreement (SK/TW)

For the South Korea and Taiwan corpora, where no human ground truth exists, "correct" is defined as agreeing with the majority-vote classification (4+ of 6 models). Responses where models assign higher token-level probability to the classification label are also responses where more models agree on that classification ($r_s = 0.46$, $p < 0.001$). Figure C.1 shows the relationship: mean predicted probability rises from 0.81 when only two models agree to 0.98 when all six agree.

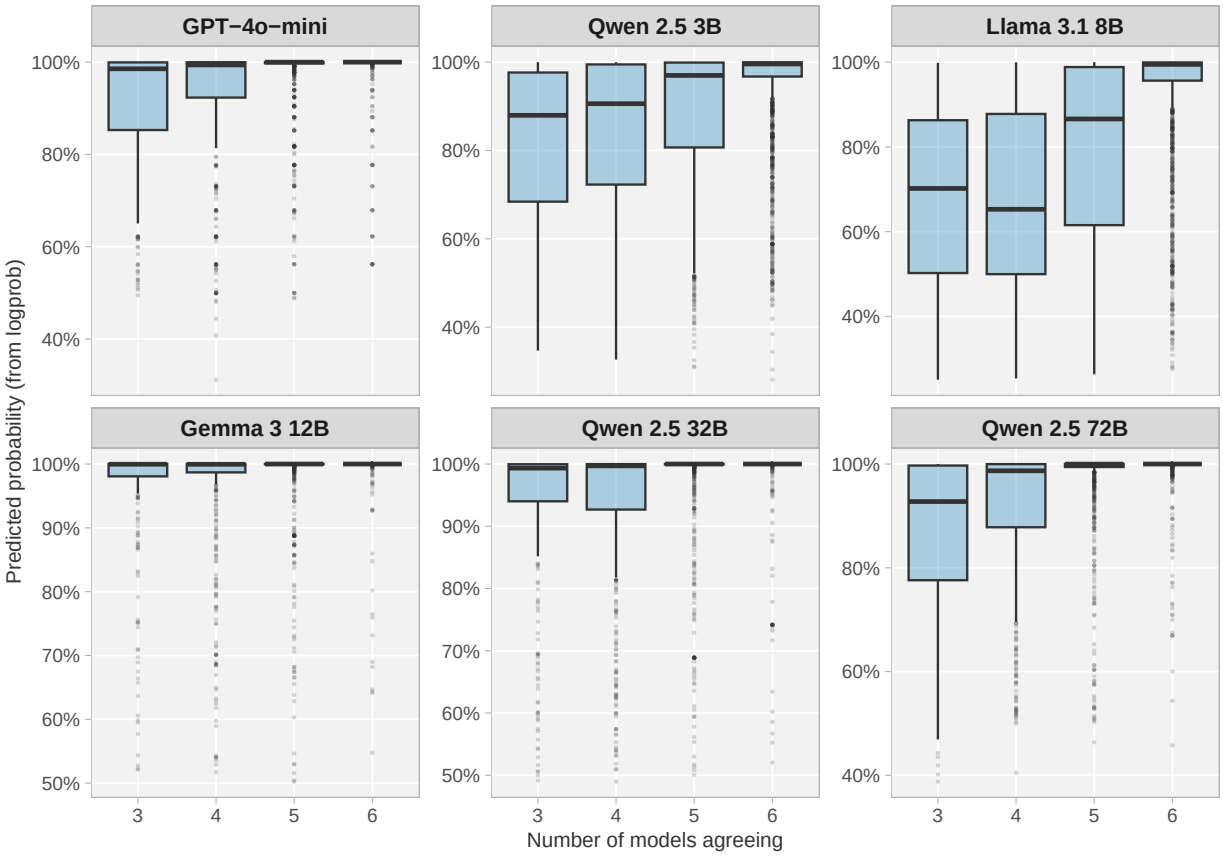

Figure C.1: Distribution of predicted probability (from token logprobs) by the number of models agreeing on the classification. Higher model certainty corresponds to broader cross-model consensus.

Calibration against human ground truth (NK entrepreneurship)

The NK entrepreneurship dataset provides the strongest calibration test because it includes adjudicated human-coded ground truth. Here, "correct" means agreement with the expert consensus code. Figure C.2 presents calibration curves for each model: the predicted probability (binned into deciles) against the observed accuracy within each bin. A perfectly calibrated model would fall along the diagonal. All six models are overconfident: even when assigning 90%+ probability, their accuracy against human codes is only 60–85%. ECE ranges from 0.145 (Qwen 2.5 72B) to 0.261 (Gemma 3 12B). Larger models are better calibrated, consistent with the model size pattern in inter-rater agreement.

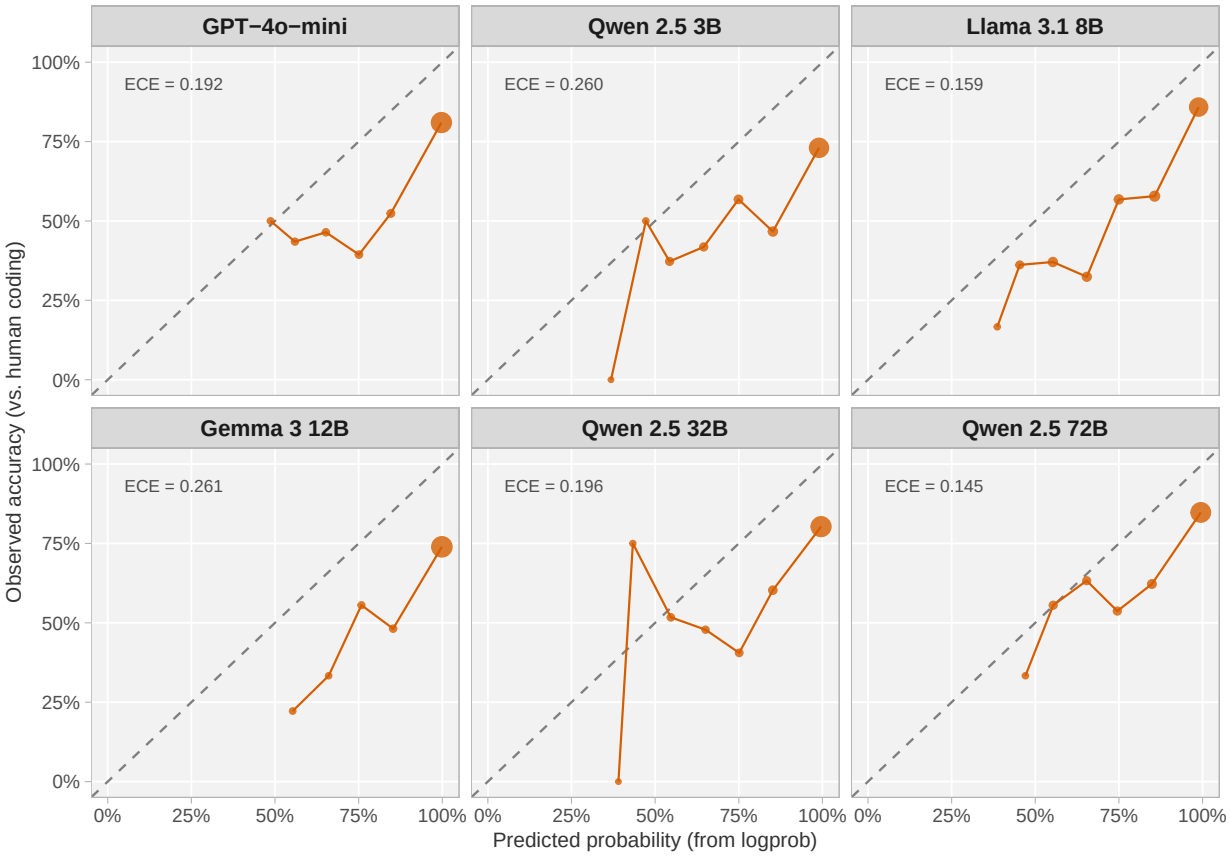

Figure C.2: Calibration curves for the N.K. entrepreneurship dataset (vs. human-adjudicated ground truth). Each panel shows one model. The dashed line is perfect calibration. ECE = Expected Calibration Error. All models fall below the diagonal: they are overconfident. Note: N.K. = North Korea.

Model	ECE (SK/TW)	ECE (N.K.)	r_s (logprob vs. n_{agree})
GPT-4o-mini	0.026	0.192	0.657
Qwen 2.5 3B	0.097	0.260	0.361
Llama 3.1 8B	0.045	0.159	0.570
Gemma 3 12B	0.084	0.261	0.640
Qwen 2.5 32B	0.038	0.196	0.621
Qwen 2.5 72B	0.012	0.145	0.637

Table C.6: Expected Calibration Error (ECE) and Spearman correlation between token-level logprobs and inter-rater agreement (n_{agree}), by model. SK/TW calibration is measured against majority vote; N.K. calibration is measured against human-adjudicated ground truth. Models appear well-calibrated against each other (low ECE for SK/TW) but overconfident relative to human coding (high ECE for N.K.). Note: N.K. = North Korea.

By contrast, Figure C.3 shows calibration against the majority vote for SK/TW, where ECE values are

much lower (0.012–0.097). This gap illustrates that inter-model agreement inflates apparent calibration: models that share pretraining data tend to be uncertain about the same responses, so the majority vote is a lenient benchmark. The NK ground-truth comparison provides the more realistic assessment.

Figure C.3: Calibration curves for the SK/TW corpus (vs. majority-vote classification). ECE values are much lower than for the NK ground-truth comparison, reflecting shared uncertainty patterns across models with overlapping pretraining data.

Self-reported confidence vs. token probability

The correlation between self-reported confidence (coded as LOW = 1, MEDIUM = 2, HIGH = 3) and the token-level probability is weak to moderate, ranging from $r_s = 0.19$ (GPT-4o-mini) to $r_s = 0.57$ (Llama 3.1 8B). This confirms that self-reported confidence ratings carry some information about actual model certainty but are not a reliable substitute for probability-based uncertainty measures. LLMs are generally not calibrated classifiers in the way that, for example, logistic regression models are calibrated by construction. The token probabilities are informative about relative confidence but should not be interpreted as true probabilities of correctness.

For practitioners, the practical implication is that token-level logprobs, when available, provide a more informative uncertainty signal than self-reported confidence ratings. Responses where the model assigns low probability to its chosen code are candidates for manual review or exclusion. In a multi-model framework, logprobs and inter-rater agreement provide complementary uncertainty estimates: logprobs measure within-model uncertainty, while agreement measures between-model uncertainty.

Validity of the logprob measure

The log-probability is computed over the model's entire vocabulary (over 100,000 tokens), not just the five valid classification codes. A natural concern is whether the logprob reflects classification confidence or merely token-level noise (e.g., competition between spelling variants like `none` and `NONE` rather than between substantive codes). Table C.7 addresses this directly. For GPT-4o-mini, the five valid code prefixes capture over 99% of the probability mass for every response. The mean probability distribution, conditional on the chosen code, shows strong diagonal dominance (96–98% on the chosen code) with the remaining mass distributed among the four competing codes. For the 9% of responses where the model assigns less than 90% probability to the chosen code, the top alternatives are competing codes, not spelling variants. This confirms that the classification prompt effectively constrains the model's predictive distribution to the codebook categories, making the logprob a valid proxy for classification confidence.

Chosen code	P(civic)	P(funct.)	P(econ.)	P(ident.)	P(none)
Civic commitment	**.963**	.018	.007	.007	.006
Functional integration	.006	**.983**	.005	.002	.003
Economic focus	.005	.009	**.980**	.002	.004
Identity concern	.007	.005	.002	**.962**	.024
None of above	.001	.009	.003	.016	**.970**

Table C.7: Mean predicted probability distribution over the five classification codes, conditional on the chosen code (GPT-4o-mini, SK/TW corpus). Rows sum to $\approx$1.0; residual mass on non-code tokens is $<$0.001 in all cases. The diagonal dominance confirms that the classification prompt concentrates the model's predictive distribution onto the valid codes, making the logprob a valid measure of classification confidence.

C.7 Per-model cue effects

Table C.8 presents the treatment effect of the social integration cue on each classification code for every model individually, without majority-vote aggregation. Each row shows the change in code prevalence

(in percentage points) when the cue is shown versus not shown, computed from that model's full set of classifications.

All six models, from the smallest (Qwen 2.5 3B, runnable on a consumer GPU) to the largest (Qwen 2.5 72B, requiring a university HPC node), agree on the direction of every substantive effect. Civic commitment increases in both countries across all models; functional integration and economic focus decrease; identity concern is stable.

Model	Country	Civic Commitment	Functional Integration	Economic Focus	Identity Concern
Proprietary					
GPT-4o-mini	South Korea	+16.1	−7.8	−5.7	−4.7
GPT-4o-mini	Taiwan	+17.9	−8.1	−10.1	−1.2
Open-weight (Qwen 2.5 family, size scaling)					
Qwen 2.5 3B	South Korea	+9.3	−3.0	−6.7	−1.4
Qwen 2.5 3B	Taiwan	+17.7	−6.8	−13.2	+1.8
Qwen 2.5 32B	South Korea	+13.1	−5.8	−4.6	−4.3
Qwen 2.5 32B	Taiwan	+16.5	−4.0	−10.9	−1.7
Qwen 2.5 72B	South Korea	+14.3	−7.0	−6.1	−3.0
Qwen 2.5 72B	Taiwan	+18.1	−6.8	−11.9	−0.6
Open-weight (other model families)					
Llama 3.1 8B	South Korea	+12.2	−7.5	−4.4	−2.6
Llama 3.1 8B	Taiwan	+15.9	−10.9	−6.5	−0.9
Gemma 3 12B	South Korea	+14.3	−6.0	−5.5	−4.6
Gemma 3 12B	Taiwan	+14.4	−4.3	−10.4	−0.7

Table C.8: Change in code prevalence (percentage points) when the social integration cue is shown, by model and country. Each model's classifications are used independently; no majority-vote aggregation is applied. The `other` category is omitted. All models agree on the direction of all substantive effects. The Qwen 2.5 family (3B, 32B, 72B) permits direct comparison of model size effects within the same architecture.

C.8 Cue effect by code (majority vote)

Table C.9 presents the prevalence of each substantive code when the social integration cue was shown versus not shown, using majority-vote classification (4+ of 6 models agree). The `other` category is excluded.

Country	Code	No Cue (%)	Cue (%)	Diff (pp)
South Korea	Civic Commitment	2.4	20.9	+18.5
South Korea	Functional Integration	44.2	35.0	−9.2
South Korea	Identity Concern	27.1	24.3	−2.8
South Korea	Economic Focus	26.3	19.8	−6.5
Taiwan	Civic Commitment	1.3	24.4	+23.1
Taiwan	Functional Integration	43.7	35.6	−8.1
Taiwan	Identity Concern	19.7	18.7	−1.0
Taiwan	Economic Focus	35.3	21.3	−14.0

Table C.9: Prevalence of each classification code by treatment condition, using majority-vote classification (4+ of 6 models agree). Differences are in percentage points. The `other` category is excluded; percentages are computed among substantive codes only.

D Framework Portability

The main analysis uses a conjoint design with within-design treatment randomization: the social integration attribute is randomly withheld, creating a clean cue-shown vs. cue-not-shown comparison that STM can incorporate as a prevalence covariate. Most conjoint experiments do not have this feature. To test whether the framework works in the more common setup, this appendix applies it to a standard conjoint on immigrant *admission* to South Korea (Denney & Green 2021), where all attributes are always shown and no within-design treatment randomization is available. The setup is lighter: LDA topic modeling (replicating the original analysis) and LLM classification with two models (GPT-4o-mini and Qwen 2.5 3B).

Denney & Green (2021) embedded open-text responses in a conjoint experiment examining South Korean attitudes toward immigrant admission ($N = 1{,}008$). Respondents evaluated paired immigrant profiles varying on seven attributes (reason for application, country of origin, language skills, profession, employment plans, gender, and ethnicity) and explained their choice in an open-text response.

The immigration attitudes literature identifies three competing explanations for how the public evaluates immigrants: sociotropic concerns (national-level norms and contribution), economic self-interest (labor market and fiscal concerns), and ethnocultural identity (origin-based preferences) (Hainmueller & Hopkins 2015). Applying the framework from the main analysis, the trichotomy is refined into five codes that distinguish functional integration (language ability, adaptation, settlement effort) from sociotropic reasoning.

Code	Definition
Sociotropic	Social harmony, community benefit, public safety, normative fit
Economic	Employment, labor market, fiscal concerns (personal or national)
Integration	Language ability, cultural adaptation, settlement effort
Identity	Ethnicity, national origin, gender, religion, co-ethnic preference
Other	Uninformative, vague impressions, refusals

Table D.1: Classification codes for the immigration admission conjoint.

Table D.2 shows the six LDA topics ($K = 6$) with revised labels informed by the five-code framework. Two topics capture functional integration (language acquisition and Korean fluency/adaptation, together 36% of documents), one captures identity (co-ethnic preference, 17%), one captures sociotropic

reasoning (societal benefit, 16%), one captures economic concerns (employment plans, 13%), and one is genuinely mixed (profession and nationality, 18%). LDA rather than STM is used because the original study did not include a treatment randomization amenable to prevalence covariates.

Topic	Label	Framework	Top terms (English gloss)	%
1	General Profile Evaluation	Mixed	occupation, immigration, ethnicity, nationality	18.0
2	Employment Plans & Purpose	Economic	job-seeking, activity, asylum, plan	13.4
3	Co-ethnic Preference	Identity	co-ethnic, employment, North Korea, person	16.8
4	Integration Effort	Integration	language, woman, settlement, effort	16.1
5	Societal Benefit	Sociotropic	Korea, country, help, society	15.8
6	Language & Employment	Integration	Korean (lang.), fluent, adaptation, ability	19.9

Table D.2: LDA topics ($K = 6$) with revised framework labels.

For LLM classification, GPT-4o-mini and Qwen 2.5 3B independently classify the 1,008 responses using the five-code codebook, achieving 79.2% agreement (Cohen's $\kappa = 0.737$). Cross-tabulating LDA topic assignments with LLM classifications shows convergence on the core constructs: for four of five non-mixed LDA topics, the dominant LLM code matches the expected framework label. Topic 2 (Employment) maps to economic, Topic 3 (Co-ethnic Preference) to identity, Topic 5 (Societal Benefit) to sociotropic, and Topic 6 (Language & Employment) to integration.

What Denney & Green (2021) described broadly as "sociotropic" reasoning decomposes into three distinct constructs (sociotropic, economic, and integration), each with a corresponding LDA topic. Functional integration emerges as the single largest category in both LDA (36% of documents) and LLM classification (20–27%), yet it was underspecified in the original analysis. Applying the framework retrospectively to existing data can sharpen constructs that were too coarse in the original study.

E Additional Robustness Analyses

E.1 Topic count selection diagnostics

Tables E.1 and E.2 report `searchK()` diagnostic metrics for the candidate topic counts evaluated in each country. $K = 6$ was selected for South Korea based on the coherence-exclusivity balance: exclusivity is past the main elbow (9.36), semantic coherence is better than $K = 7$ or $K = 8$, and residuals are substantially lower than $K = 4$ or $K = 5$. $K = 9$ produced a degenerate solution (2 EM iterations) and was excluded. $K = 8$ was selected for Taiwan because it achieves the best semantic coherence (-228.5, the least negative across all K) while exclusivity (9.56) is well past the elbow, and residuals improve substantially from $K = 7$ (5.57) to $K = 8$ (5.08).

K	Exclusivity	Sem. Coherence	Held-Out Lik.	Residuals	ELBO
4	9.025	-257.8	-4.671	4.635	$-17{,}264$
5	9.275	-264.9	-4.625	4.521	$-17{,}266$
6	**9.362**	$\mathbf{-274.2}$	-4.626	**4.360**	$-17{,}204$
7	9.446	-278.3	-4.653	4.220	$-17{,}192$
8	9.483	-282.6	-4.550	3.866	$-17{,}103$
10	9.614	-281.4	-4.632	3.675	$-17{,}091$
12	9.661	-272.8	-4.597	4.054	$-17{,}055$

Table E.1: South Korea `searchK()` diagnostics ($K = 4$–12). $K = 9$ omitted (degenerate solution). Selected $K = 6$ (bold).

K	Exclusivity	Sem. Coherence	Held-Out Lik.	Residuals	ELBO
5	9.293	-230.0	-4.537	5.653	$-24{,}893$
6	9.474	-244.0	-4.450	5.629	$-24{,}617$
7	9.515	-242.2	-4.592	5.572	$-24{,}743$
8	**9.555**	$\mathbf{-228.5}$	-4.489	**5.084**	$-24{,}358$
9	9.613	-249.5	-4.460	5.011	$-24{,}347$
10	9.615	-256.1	-4.537	4.882	$-24{,}563$
12	9.701	-251.6	-4.421	5.431	$-24{,}312$
15	9.749	-277.1	-4.457	6.047	$-24{,}094$

Table E.2: Taiwan `searchK()` diagnostics ($K = 5$–15). Selected $K = 8$ (bold).

E.2 Majority-vote threshold sensitivity

Table E.3 shows the civic commitment treatment effect under alternative majority-vote thresholds. Effects are stable across thresholds (14.3–15.3 pp in South Korea, 17.5–18.2 pp in Taiwan). Exclusions are balanced across treatment conditions at all thresholds (all χ^2 $p > 0.20$). Dropping GPT-4o-mini entirely and using a 3-of-5 open-weight majority produces near-identical estimates.

Threshold	Country	No Cue (%)	Cue (%)	Effect (pp)	N incl.
3/6 ($\geq$3)	South Korea	3.8	18.1	$+14.3$	2,007
3/6 ($\geq$3)	Taiwan	1.8	19.7	$+17.9$	2,015
4/6 ($\geq$4)	South Korea	2.0	16.7	$+14.7$	1,820
4/6 ($\geq$4)	Taiwan	1.0	19.2	$+18.2$	1,935
5/6 ($\geq$5)	South Korea	1.0	16.3	$+15.3$	1,569
5/6 ($\geq$5)	Taiwan	0.6	18.3	$+17.7$	1,693
3/5 no GPT	South Korea	2.6	16.8	$+14.3$	1,953
3/5 no GPT	Taiwan	1.3	18.8	$+17.5$	1,991

Table E.3: Civic commitment treatment effect under alternative majority-vote thresholds. The "3/5 no GPT" row excludes GPT-4o-mini and uses a 3-of-5 open-weight majority. Exclusions are balanced across treatment conditions at all thresholds.

E.3 Keyword baseline comparison

To contextualize the LLM contribution, a simple keyword/dictionary classifier was applied to all 4,049 responses. The dictionary maps Korean and Chinese keywords to the five classification codes (e.g., □□/□□ → civic_commitment; □□/□□ → economic_focus; □□□/□□ → functional_integration; □□/□□ → identity_concern). Responses matching no keywords are classified as "other."

When a response matches keywords from multiple categories, the first match in code priority order (civic, functional, identity, economic) is assigned. The keyword classifier achieves 62% agreement with the LLM majority vote (Cohen's $\kappa = 0.51$, moderate). The main limitation is coverage: 51% of responses receive no keyword match and default to "other," compared to 21% for the LLM. The keyword baseline detects the civic commitment treatment effect in the correct direction ($+5.9$ pp in South Korea, $+7.6$ pp in Taiwan) but substantially attenuates the magnitude relative to the LLM ($+14.7$ and $+18.2$ pp). The LLM's value-added is greatest for short, semantically ambiguous responses where context and world

knowledge compensate for the absence of explicit keywords.

E.4 Treatment effects by response length

Table E.4 reports the civic commitment treatment effect separately for short ($\leq$ 7 characters, below the corpus median) and long ($\geq$ 8 characters) responses. The treatment effect is present and directionally consistent across all length groups but is substantially larger for longer responses. Even responses of five characters or fewer show a +7 pp civic commitment shift, confirming that the treatment effect is not an artifact of long, elaborated responses.

Country	Length	No Cue (%)	Cue (%)	Effect (pp)	N
South Korea	Long ($\geq$8 chars)	3.1	21.5	+18.4	1,099
South Korea	Short ($\leq$7 chars)	0.3	9.3	+9.0	718
Taiwan	Long ($\geq$8 chars)	2.6	34.4	+31.8	703
Taiwan	Short ($\leq$7 chars)	0.2	9.9	+9.7	1,235

Table E.4: Civic commitment treatment effect by response length (median split at 7 characters). Effects are present in both length groups but larger for longer responses, consistent with longer responses providing more classifiable signal.

E.5 Prompt templates

The system prompt is identical across all six classifiers and is generated from the YAML codebook. It consists of four sections: (1) task framing and instructions for handling short responses and the language/residence boundary; (2) five classification codes with definitions, inclusion criteria, and exclusion criteria; (3) 25 labeled few-shot examples in Korean and Traditional Chinese; and (4) nine classification rules and the JSON output format.

The prompt opens with task framing:

> "You are a research assistant coding open-ended survey responses about attitudes toward immigrant naturalization in Taiwan and South Korea. [...] Your task is to classify each response."

Two critical instructions follow: (1) short responses (1–3 words) must be classified if they contain any substantive content; brevity alone is never a reason for the residual code; (2) standalone mentions of language ability or residence duration map to `functional_integration`, not `civic_commitment`, un-

less explicitly framed with civic engagement language. The five code definitions (with inclusion/exclusion criteria) and 25 labeled few-shot examples in Korean and Traditional Chinese follow, along with nine numbered classification rules and the JSON output format. The full 168-line prompt is reproduced in the replication archive at `output/classification/system_prompt.txt`.

The user message differs between classifiers. For GPT-4o-mini, the user message is: "Classify this survey response: [text]." For the open-weight models via Ollama, the user message adds an explicit enumeration of valid codes and a JSON output format reminder: "The 'code' MUST be exactly one of: civic_commitment, functional_integration, identity_concern, economic_focus, none_of_above. Respond with ONLY a JSON object." The additional constraint reduces parse errors for smaller models that occasionally invent code names or wrap JSON in markdown.